

# Optimal Sizing and Operation of Battery Energy Storage Systems in Renewable Energy Systems Using Multi-Objective Optimization and Artificial Intelligence

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## Abstract

The integration of battery energy storage systems (BESS) provides a viable solution to address the uncertainty and the intermittent nature challenges of renewable energy generation. However, determining the optimal size and operation of BESS remains complex due to the trade-offs between total cost, self-sufficiency, system reliability, and BESS lifespan. Existing studies often focus on single-objective frameworks, overlooking the potential benefits of combining artificial intelligence (AI) for prediction and control. This paper presents a novel framework that combines multi-objective optimization with AI prediction techniques to determine the optimal size and operation of BESS in grid-connected microgrids with photovoltaic (PV) generation. Six different multi-objective optimization models are developed, incorporating both economic and technical criteria, including system cost, BESS cycle life, system reliability, and self-sufficiency. To improve prediction accuracy, XGBoost and LSTM are utilized for predicting PV generation and energy demand. Additionally, an AI-driven algorithm is proposed to schedule BESS charging and discharging based on forecast data. Comprehensive computational experiments demonstrate the effectiveness of the proposed framework and its potential for determining the optimal size, power rating, operation, and scheduling of BESS.

**Keywords:** Battery Energy Storage System, Multi-Objective Optimization, Artificial Intelligence Techniques, Optimal Sizing, Renewable Energy, Energy Management.

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## Nomenclature

| Model Indices         |                                                                                                     |
|-----------------------|-----------------------------------------------------------------------------------------------------|
| $i$                   | Index of PV ( $i = 1, \dots, N_{PV}$ ).                                                             |
| $k$                   | Index of BESS ( $k = 1, \dots, N_{BESS}$ ).                                                         |
| $s$                   | Index of scenarios                                                                                  |
| Model Parameters      |                                                                                                     |
| $AM_0$                | The air mass under standard test conditions, which is 1.5 [AM1.5].                                  |
| $AM_t$                | The air mass at the time $t$ [AM].                                                                  |
| $Budget_{BESS}$       | The budget limit for BESS [€].                                                                      |
| $Cost_{BESS_k, kW}$   | Power rating cost of $k^{th}$ BESS device [€/kW].                                                   |
| $Cost_{BESS_k, kWh}$  | Capacity rating cost of $k^{th}$ BESS device [€/kWh].                                               |
| $Cost_{PV_i, kW}$     | Power rating cost of $i^{th}$ PV [€/kW].                                                            |
| $CE_{BESS_k}^{max}$   | The upper limit value of charging energy of $k^{th}$ BESS device [kWh].                             |
| $CE_{BESS_k}^{min}$   | The lower limit value of charging energy of $k^{th}$ BESS device [kWh].                             |
| $DCC_t$               | Direct consumer costs for not supplied energy at time $t$ [€/kWh].                                  |
| $DOD_{BESS_k}$        | Depth of discharge (DOD) of the $k^{th}$ BESS device [%].                                           |
| $DP_{BESS_k}^{max}$   | The upper limit value of discharging power of $k^{th}$ BESS device [kW].                            |
| $DP_{BESS_k}^{min}$   | The lower limit value of discharging power of $k^{th}$ BESS device [kW].                            |
| $\Delta t$            | The time interval [hour].                                                                           |
| $ES_{BESS_k}^{max}$   | The maximum energy capacity (energy storage) of $k^{th}$ BESS device [kWh].                         |
| $ES_{BESS_k}^{min}$   | The minimum energy capacity (energy storage) of $k^{th}$ BESS device [kWh].                         |
| $IA_{PV_i}$           | The installation area of the $i^{th}$ PV [ $m^2$ ].                                                 |
| $ISCO_t$              | Indirect social costs for not supplied energy at hour $t$ [€/kWh].                                  |
| $N_{BESS}$            | The number of BESS.                                                                                 |
| $N_{PV}$              | The number of PV.                                                                                   |
| $\eta C_{BESS_k}$     | The charging efficiency of $k^{th}$ BESS device [-].                                                |
| $\eta D_{BESS_k}$     | The discharging efficiency of $k^{th}$ BESS device [-].                                             |
| $\eta RT_{BESS_k}$    | The energy round trip efficiency of the BESS [%].                                                   |
| $\eta_{PV_i t}$       | Efficiency of the $i^{th}$ PV [-].                                                                  |
| $OMC_{BESS_k}$        | Operation and maintenance cost of $k^{th}$ BESS [€/kWh].                                            |
| $OMC_{BESS_k, kW}$    | Operation and maintenance cost of $k^{th}$ BESS's power [€/kW].                                     |
| $OMC_{BESS_k, kWh}$   | Operation and maintenance cost of $k^{th}$ BESS's capacity [€/kWh].                                 |
| $OMC_{PV_i}$          | Operation and maintenance cost of $i^{th}$ PV [€/kWh].                                              |
| $PLR_{BESS_k}$        | The loss (self-discharge) rate of $k^{th}$ BESS device [-].                                         |
| $Serv\_Life_{BESS_k}$ | Service life of $k^{th}$ BESS device [times].                                                       |
| $SOC_{BESS_k}^{max}$  | The upper limit value of the state of charge (SOC) of the $k^{th}$ BESS device [%].                 |
| $SOC_{BESS_k}^{min}$  | The lower limit value of the state of charge (SOC) of the $k^{th}$ BESS device [%].                 |
| $SRI_0$               | The solar radiation intensity under standard test conditions, which is 1000 [ $W/m^2$ ].            |
| $SRI_{PV_i t}$        | The solar radiation intensity (SRI) over the period $t$ [ $Wh/m^2$ ].                               |
| $t$                   | The time unit [hour].                                                                               |
| $T_0$                 | The outdoor ambient temperature under standard test conditions, which is 25 °C.                     |
| $T_{out t}$           | The outdoor ambient temperature at time $t$ [°C].                                                   |
| $TES^{max}$           | The maximum energy capacity of the transmission line over the period $t$ [kWh].                     |
| $TES^{min}$           | The minimum energy capacity of the transmission line over the period $t$ [kWh].                     |
| $UC_{Sup t}$          | The unit cost of purchasing electric energy from the power grid [€/kWh].                            |
| $UINC_t$              | Unit income of electric energy sales [€/kWh].                                                       |
| Vul                   | The vulnerability factor of the consumer [%].                                                       |
| Model Variables       |                                                                                                     |
| $E_{PV t}$            | Total active energy generation from PV at time $t$ [kWh].                                           |
| $E_{PV_i t}$          | Active Energy generation from $i^{th}$ PV at time $t$ [kWh].                                        |
| $P_{PV_i}^{rated}$    | Rated power of $i^{th}$ PV panel [kW].                                                              |
| $\beta_t^{TPS}$       | Binary variable related to the selling surplus power generation to the power grid at the time $t$ . |
| $Cost_{BESS\_DOD}$    | The cost per cycle of BESS related to the DOD [€].                                                  |
| $Cost_{BESS\_OM}$     | The total BESS operation and maintenance costs [€].                                                 |
| $Cost_{BESS_k\_OM t}$ | Operation and maintenance costs of $k^{th}$ BESS device at time $t$ [€].                            |

|                              |                                                                                                    |
|------------------------------|----------------------------------------------------------------------------------------------------|
| $Cost_{Cap\_BESS}$           | The total daily capital cost of BESS [€].                                                          |
| $Cost_{Cap\_PV}$             | The total daily capital cost of PV panels [€].                                                     |
| $Cost_{DEP\_PV}$             | The total daily depreciation cost of PV panels [€/days].                                           |
| $Cost_{ENS\ t}$              | The cost of the energy not supplied in the system at time $t$ [€].                                 |
| $Cost_{PV\_OM}$              | The total PV operation and maintenance costs [€].                                                  |
| $Cost_{PV\_i\_OM\ t}$        | Operation and maintenance costs of $i^{th}$ PV panel at time $t$ [€].                              |
| $Cost_{Total\_Cap}$          | The total capital cost [€].                                                                        |
| $Cost_{Total\_DEC}$          | The total daily energy cost supplied from the power grid [€].                                      |
| $Cost_{Total\_DEP}$          | The total depreciation cost [€].                                                                   |
| $Cost_{Total\_ENS}$          | The total cost of the energy not supplied [€].                                                     |
| $Cost_{Total\_OM}$           | The total operation and maintenance cost [€].                                                      |
| $CE_{BESS\ t}$               | Total charging energy in BESSs over the period $t$ [kWh].                                          |
| $CE_{BESS_k\ t}$             | Charging energy of $k^{th}$ BESS device over the period $t$ [kWh].                                 |
| $CE_{PG\ t}$                 | Total electric energy for charging BESSs supplied from the power grid at hour $t$ [kWh].           |
| $CE_{PV\ t}$                 | Total energy for charging BESSs supplied from PV at hour $t$ [kWh].                                |
| $DE_{BESS_k\ t}$             | Discharging energy from BESSs over the period $t$ [kWh].                                           |
| $DE_{BESS\ t}$               | Total discharging energy from BESSs at hour $t$ [kWh].                                             |
| $DP_{BESS_k}^{rated}$        | Rated power of $k^{th}$ BESS device [kW].                                                          |
| $ES_{BESS\ t}$               | Total energy storage (total energy capacity) in BESSs at hour $t$ [kWh].                           |
| $ES_{BESS_k\ t}$             | Energy storage (energy capacity) in the $k^{th}$ BESS device at hour $t$ [kWh].                    |
| $ES_{BESS_k}^{rated}$        | Rated energy storage capacity (energy storage) of $k^{th}$ BESS device [kWh].                      |
| $\xi_{BESS_k\ t}^{charg}$    | Binary variable related to charging status of the $k^{th}$ BESS device at hour $t$ [-].            |
| $\xi_{BESS_k\ t}^{discharg}$ | Binary variable related to discharging status of the $k^{th}$ BESS device at hour $t$ [-].         |
| $Income_{Total}$             | The total daily income [€].                                                                        |
| $Cycle\_Life_{BESS_k}$       | Lifecycle of $k^{th}$ BESS device [times].                                                         |
| $E_{NS\ t}$                  | Total energy not supplied in the system at hour $t$ [kWh].                                         |
| $Serv\_Life_{PV_i}$          | Service life of $i^{th}$ PV panel [days].                                                          |
| $SOC_{BESS_k\ t}$            | The state of charge of the $k^{th}$ BESS device at hour $t$ [%].                                   |
| $TBLD_{PG\ t}$               | Total building load demand supplied from the power grid at hour $t$ [kWh].                         |
| $TBLD_{PV\ t}$               | Total building load demand supplied from PV at hour $t$ [kWh].                                     |
| $TBLD_t$                     | Total building load demand at hour $t$ [kWh].                                                      |
| $TLD^{mean}$                 | The average value of load demand within a day [kWh].                                               |
| $TLD_t$                      | Total load demand at hour $t$ [kWh].                                                               |
| $TES_{PG\ t}$                | Total electric energy supplied from the power grid at hour $t$ [kWh].                              |
| $TES_t$                      | Total energy supply at hour $t$ [kWh].                                                             |
| $TSE_{PV\ t}$                | Total surplus electric energy generation from PV that is injected into the grid at hour $t$ [kWh]. |

## 1. Introduction

Wind and solar energy are among the most important renewable energy sources (RES) and play a crucial role in reducing reliance on fossil fuels and mitigating carbon emissions [1]. Despite their advantages, RES introduces several challenges for power systems. Wind and solar energy generation are highly dependent on weather conditions, which can cause instability in power systems [1, 2]. On the other hand, conventional power systems often struggle to adapt to renewable energy generation. For example, excessive photovoltaic (PV) generation occurs during the afternoon, which may result in lower net demand, while PV cannot meet the higher net demand in the morning and evening [1].

To address these challenges, energy storage systems (ESS) have been widely proposed to improve the flexibility and reliability of power systems with RES integration [1, 3]. ESSs can be battery energy storage systems (BESS), thermal storage systems (TS), hydrogen storage systems (HS), etc., depending on the desired application and performance [3, 4]. Due to significant advantages, BESSs have been widely adopted in power systems that integrate RES.

Among these advantages are fast response, controllability, peak load management, availability, and a wide range of applications to enhance short-term power quality, long-term energy management, reliability, and power supply [5, 6]. However, their relatively high cost and limited lifespan make the optimal sizing and operation of BESS a complex problem. Oversizing can lead to unjustified costs, and under-sizing may result in unmet requirements [1, 3].

In determining the optimal size of BESS, various power system requirements are considered. These requirements can be generally divided into four categories: financial, technical, environmental, and hybrid [1, 3]. Renewable energy systems are categorized into four types: distributed renewable energy systems (grid-connected), microgrids (MGs, operating in two modes: grid-connected and islanded), standalone hybrid renewable energy systems (islanded), and power plant systems (grid-connected) [1].

The literature offers a variety of methods for BESS sizing and operation planning, including probabilistic, analytical, heuristic, artificial intelligence (AI)-based, and mathematical optimization approaches [1, 3]. The following is a review of methods proposed in the literature for optimizing the size of the BESS.

In [7], the energy-to-power ratio is defined as a coefficient, and a model is proposed to enable quick evaluation of the feasibility of BESS for commercial buildings. In [8], a single-objective optimization model is proposed to enhance the reliability of an MG integrated with RES. The objective function is established to minimize the total cost of the MG. A genetic algorithm-based method for measuring the energy storage system in MGs has been proposed in [9], which aims to determine the energy and power capacities of the storage system to minimize the operating cost of the MG. In [10], a BESS capacity planning model is proposed to address the problem of peak and load shaving. An optimal BESS sizing method for a smart MG with PV systems and air conditioning resources is proposed in [11]. Since balancing energy demand in an isolated MG is a critical issue, especially in the presence of RES, a multi-year operational planning model is proposed in [12] to determine the optimal power and energy capacity of BESS. In [13], the effect of BESS in increasing self-consumption and independence in homes fitted with PV panels has been investigated. They evaluated economic feasibility and potential environmental impacts by determining the most effective BESS for maximizing solar energy utilization. In [14], the effect of BESS on increasing the self-consumption of electricity produced by PV has been investigated. The proposed model maximizes self-sufficiency and, thus, the level of PV power consumption by minimizing the amount of electricity purchased from the grid.

In [15], the authors proposed a mathematical model that incorporates technical and economic metrics for grid-connected systems. Their study demonstrated that integrating BESS enhances self-consumption and self-sufficiency rates. A study on energy and financial flows in apartments equipped with PV systems and BESS was conducted in [16], revealing that central batteries could improve solar self-consumption, increase self-sufficiency, and reduce peak demand. An economic analysis of PV systems with BESS was performed in [17], where a mathematical model identified the break-even point at which PV battery systems become economically viable. In [18], a methodology for optimizing the sizes of PV and electrochemical battery systems was proposed to maximize self-sufficiency and cost-effectiveness in grid-

connected microgrids. Their findings indicated that PV cost-effectiveness was high enough to justify the installation of expensive batteries.

A battery capacity configuration method aimed at improving the self-sufficiency rate (SSR) and self-consumption rate (SCR) in PV systems was introduced in [19]. The results showed a significant increase in SSR and SCR with larger battery capacities. Similarly, in [20], a method for configuring battery capacity was presented. The study revealed that SSR is more sensitive to PV capacity than SCR and that a centralized storage scheme generally outperforms a distributed one.

Recent research highlights AI-based advances in BESS sizing. In [21, 22], a study was conducted on the sizing and management of energy storage systems using price control and artificial intelligence (AI). The results indicated that as the size of the BESS increased, the profit margin also grew significantly. In [23], a load forecasting technique based on the Long Short-Term Memory (LSTM) method was used to analyze the appropriate size of BESS, which included residential and electric vehicle (EV) charging demand. In [24], the importance of AI advancements in EV energy storage systems is highlighted, focusing on battery management systems (BMS). In [25], the effectiveness of a two-stage neural network-based model predictive control (MPC) algorithm for energy consumption optimization and peak demand charges management using BESS is investigated. The attentive convolutional deep reinforcement learning (AC-DRL) method is proposed in [26] for real-time electricity markets, and a viable strategy for BESS is developed to unlock economic potential by playing a backup role to reduce solar power curtailments while participating in the electricity market.

Mathematical optimization techniques have been widely employed in the literature for determining the optimal size of BESSs. Table 1 summarizes the details of the optimization models proposed in the literature. This table displays the type of optimization problem (single-objective or multi-objective), the type of MG system (connected or isolated), the type of BESS (lithium-ion or lead-acid), the type of renewable energy technology used in the system (photovoltaic PV or wind turbine WT systems), the objective functions, and the model constraints. The most commonly used objective functions in optimization models are: Minimizing the total cost (F1), Minimizing the daily load variance (F2), Maximizing the internal rate of return (F3), Minimizing the loss of power supply probability (LPSP) or unmet load (F4), Minimizing environmental pollution and emissions (F5), Maximizing the market profit or net profits (F6), Maximizing BESS benefit or life span (F7), and Minimizing energy imported from the grid or Maximizing PV consumptive rate (F8). The optimization model constraints are divided into five categories: power grid constraints (such as power balance, power factor, active and reactive demand, current, voltage, voltage regulator, and switched capacitor) (C1), PV constraints (such as power generation capacity) (C2), WT constraints (such as power generation capacity) (C3), BESS constraints (such as energy capacity, power capacity) (C4), and emission constraints (C5).

**Table 1.** Comparative analysis of the present work and existing studies.

| Ref               | Model Type       |                 | MG Type   |          | BESS Type   |           | RES Type |    | Objective Function |    |    |    |    |    |    |    | Model Constraints |    |    |    |    |
|-------------------|------------------|-----------------|-----------|----------|-------------|-----------|----------|----|--------------------|----|----|----|----|----|----|----|-------------------|----|----|----|----|
|                   | Single Objective | Multi-Objective | Connected | Isolated | Lithium-ion | Lead-acid | PV       | WT | F1                 | F2 | F3 | F4 | F5 | F6 | F7 | F8 | C1                | C2 | C3 | C4 | C5 |
| [3]               | ✓                |                 | ✓         |          | ✓           |           | ✓        |    | ✓                  |    |    |    |    |    |    |    | ✓                 | ✓  |    | ✓  |    |
| [8]               | ✓                |                 | ✓         |          |             | ✓         | ✓        | ✓  | ✓                  |    |    |    |    |    |    |    | ✓                 | ✓  | ✓  | ✓  |    |
| [9]               | ✓                |                 | ✓         | ✓        |             | ✓         |          | ✓  | ✓                  |    |    |    |    |    |    |    | ✓                 |    | ✓  | ✓  |    |
| [10]              | ✓                |                 | ✓         |          |             |           |          |    |                    | ✓  |    |    |    |    |    |    | ✓                 |    |    | ✓  |    |
| [11]              | ✓                |                 | ✓         | ✓        | ✓           |           | ✓        |    | ✓                  |    |    |    |    |    |    |    | ✓                 | ✓  |    | ✓  |    |
| [12]              | ✓                |                 |           | ✓        |             | ✓         | ✓        |    | ✓                  |    |    |    |    |    |    |    | ✓                 | ✓  |    | ✓  |    |
| [14]              | ✓                |                 | ✓         |          | ✓           |           | ✓        |    | ✓                  |    |    |    |    |    |    |    | ✓                 | ✓  |    | ✓  |    |
| [15]              | ✓                |                 | ✓         |          | ✓           |           | ✓        |    | ✓                  |    |    |    |    |    |    |    | ✓                 | ✓  |    | ✓  |    |
| [21]              | ✓                |                 | ✓         |          | ✓           |           | ✓        |    |                    |    | ✓  |    |    |    |    |    | ✓                 | ✓  |    | ✓  |    |
| [27]              | ✓                |                 | ✓         |          | ✓           |           | ✓        |    | ✓                  |    |    |    |    |    |    |    | ✓                 | ✓  |    | ✓  |    |
| [28]              |                  | ✓               | ✓         |          | ✓           |           | ✓        |    | ✓                  |    |    | ✓  | ✓  |    |    |    | ✓                 | ✓  |    | ✓  | ✓  |
| [29]              | ✓                |                 |           | ✓        | ✓           |           | ✓        |    |                    | ✓  |    |    |    |    |    |    | ✓                 | ✓  |    | ✓  |    |
| [30]              | ✓                |                 |           | ✓        | ✓           |           | ✓        |    | ✓                  |    |    |    |    |    |    |    | ✓                 | ✓  |    | ✓  |    |
| [31]              | ✓                |                 | ✓         |          | ✓           | ✓         | ✓        | ✓  | ✓                  |    |    |    |    |    |    |    | ✓                 | ✓  | ✓  | ✓  |    |
| [32]              | ✓                |                 | ✓         |          | ✓           |           | ✓        | ✓  | ✓                  |    |    |    |    |    |    |    | ✓                 | ✓  | ✓  | ✓  |    |
| [33]              | ✓                |                 | ✓         |          |             | ✓         | ✓        | ✓  | ✓                  |    |    |    |    |    |    |    | ✓                 | ✓  | ✓  | ✓  |    |
| [34]              |                  | ✓               |           | ✓        |             | ✓         | ✓        | ✓  | ✓                  |    |    | ✓  |    |    |    |    | ✓                 | ✓  | ✓  | ✓  |    |
| [35]              |                  | ✓               |           | ✓        |             | ✓         | ✓        | ✓  | ✓                  |    |    | ✓  | ✓  |    |    |    | ✓                 | ✓  | ✓  | ✓  | ✓  |
| [36]              |                  | ✓               |           | ✓        |             | ✓         | ✓        | ✓  | ✓                  |    |    | ✓  | ✓  |    |    |    | ✓                 | ✓  | ✓  | ✓  | ✓  |
| [37]              | ✓                |                 |           | ✓        |             | ✓         | ✓        | ✓  | ✓                  |    |    |    |    |    |    |    | ✓                 | ✓  | ✓  | ✓  |    |
| [38]              | ✓                |                 |           | ✓        | ✓           |           | ✓        | ✓  | ✓                  |    |    |    |    | ✓  |    |    | ✓                 | ✓  | ✓  | ✓  |    |
| [39]              |                  | ✓               | ✓         |          | ✓           |           | ✓        | ✓  | ✓                  |    |    |    |    |    | ✓  |    | ✓                 | ✓  | ✓  | ✓  |    |
| [40]              |                  | ✓               | ✓         |          | ✓           |           | ✓        |    | ✓                  |    |    |    |    |    |    | ✓  | ✓                 | ✓  |    | ✓  |    |
| [41]              |                  | ✓               | ✓         |          |             |           | ✓        | ✓  | ✓                  |    |    |    |    |    | ✓  | ✓  | ✓                 | ✓  | ✓  | ✓  |    |
| [42]              |                  | ✓               | ✓         |          | ✓           | ✓         | ✓        |    | ✓                  | ✓  |    |    |    |    |    |    | ✓                 | ✓  |    | ✓  |    |
| [43]              |                  | ✓               | ✓         |          | ✓           |           | ✓        |    | ✓                  | ✓  |    | ✓  |    |    |    |    | ✓                 | ✓  |    | ✓  |    |
| [44]              |                  | ✓               | ✓         |          | ✓           |           | ✓        |    |                    |    |    |    |    | ✓  |    | ✓  | ✓                 | ✓  |    | ✓  |    |
| <b>This study</b> |                  | ✓               | ✓         |          | ✓           |           | ✓        |    | ✓                  |    |    | ✓  | ✓  | ✓  | ✓  | ✓  | ✓                 | ✓  |    | ✓  |    |

As shown in Table 1, most existing models focus on single-objective optimization problems. Among the multi-objective models, most incorporate two objective functions. In both single-objective and multi-objective frameworks, minimizing the total system cost is commonly defined as a primary objective. The predominant system type studied is the grid-connected MG integrated with PV panels. Lithium-ion batteries are the most frequently used technology for BESS across these studies. Given the presence of conflicting objectives in determining the optimal size of BESS, where improving one objective often leads to the deterioration of another, a multi-objective optimization approach seems more appropriate and effective.

This study presents several multi-objective optimization models for a grid-connected MG that partially fulfills its energy demand through PV generation and incorporates a lithium-ion BESS. The proposed models incorporate objective functions F1 and F4–F8, either directly or indirectly, to capture a range of economic and technical performance criteria.

Batteries store and convert energy through electrochemical reactions, enabling the transformation between electrical and chemical energy. Common battery technologies used for energy storage include lithium-ion, lead-acid, flow, and sodium-sulfur batteries [27]. Table 2 summarizes the key features, advantages, and disadvantages of each battery type [22].

**Table 2.** Features, advantages, and disadvantages of various battery types [22, 27].

| Battery Type    | Advantages                                                                                       | Disadvantages                                                                            |
|-----------------|--------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|
| Lithium-ion     | Relatively mature technology, long service life, good performance, good economy, large capacity. | Safety performance, intolerant to overcharging and discharge.                            |
| Lead-acid       | Low cost, high safety, recycling options, high rate.                                             | Low energy density, short cycle life, environmental pollution.                           |
| Flow (vanadium) | High safety, high cycle times, low self-discharge rate.                                          | Low energy density, high costs.                                                          |
| Sodium-sulfur   | High energy density, long service life, no self-discharge.                                       | Running at high temperatures, Issues regarding structural design and material selection. |

According to Table 2, while lead-acid batteries are the most cost-effective option among various battery types, their lower energy density and limited charge/discharge cycles make lithium-ion batteries a more suitable and attractive choice for residential applications [22]. In this study, the lithium-ion battery was selected as the BESS technology. Table 3 presents key parameters commonly associated with electrochemical energy storage in lithium-ion batteries.

**Table 3.** Mainstream parameters of Lithium-Ion batteries [22].

| Performance           |                            |                      |                          |                |                            |
|-----------------------|----------------------------|----------------------|--------------------------|----------------|----------------------------|
| Operating voltage (V) | Energy density (Wh per kg) | Cycle life (minutes) | Energy cost (\$ per kWh) | Efficiency (%) | Operating temperature (°C) |
| 3.3~3.7               | 130~200                    | 2500~5000            | 170~400                  | 85~98          | low temperature            |

This literature review highlights a significant research gap: the absence of comprehensive, AI-enhanced multi-objective frameworks that jointly optimize both BESS sizing and forecast-based operational scheduling in grid-connected microgrids. Despite extensive research, most studies focus on single-objective optimization models that typically aim to minimize cost or improve reliability [e.g., 3, 8, 9, 10, 11, 12, 14, 15, 21, 27, 29, 30, 31, 32, 33, 37, 38], while overlooking a broader set of interrelated performance criteria. In addition, although accurate short-term forecasting of PV generation and load demand can significantly improve BESS scheduling [e.g., 21, 22], existing research rarely integrates AI prediction techniques into both sizing and control [e.g., 23, 24, 25, 26]. Even fewer studies link forecast-driven control strategies with the simultaneous optimization of system cost, BESS cycle life, reliability, and self-sufficiency [e.g., 8, 10, 15, 28, 34, 35, 36, 39, 40, 41, 42, 43, 44]. The main innovations and contributions of this work are outlined as follows:

- Development of six distinct multi-objective optimization models, incorporating total cost, BESS cycle, PV depreciation, self-sufficiency, and system reliability.
- Integration of AI prediction techniques (XGBoost and LSTM) to generate accurate forecasts of PV generation and load demand, thereby supporting BESS operators in making informed decisions.
- Design of a forecast-driven BESS scheduling algorithm, enabling adaptive charging/discharging based on real-time forecast inputs, utilizing predicted PV output and energy consumption to optimize storage behavior over a 24-hour horizon dynamically.

The remainder of the paper is organized as follows: Section 2 outlines the materials and methods. Section 3 presents the results and discussion. Computational experiments are detailed in Section 4. Finally, the conclusions are summarized in Section 5.

## **2. Materials and Methods**

This section outlines the general framework of our study and provides an overview of relevant AI techniques, along with the performance metrics used to evaluate their predictive accuracy.

### **2.1. General Model**

The primary objective of this research is to develop several multi-objective optimization models to determine the optimal capacity and power rating of BESS, accounting for the intermittency of renewable energy sources. Additionally, an optimal energy management strategy is proposed to enhance the reliability of the electricity supply by integrating renewable energy and BESS and evaluating its performance based on AI predictions of energy consumption and solar production. An AI controller is integrated into the system to support the achievement of the defined objectives. The tasks of this AI controller include:

- Prediction of load consumption;
- Prediction of energy generation from PV panels;
- Optimizing BESS operation by:
  - ✓ Predicting optimal charging and discharging periods;



- ✓ Estimating stored energy levels to enhance battery lifespan.

In this study, a grid-connected MG system comprises PV panels, BESS, DC-AC converters, load demand, and a connection to the external power grid. An overview of the main system components is illustrated in Fig. 1.

The PV system supplies electricity to both the household load and the BESS. In addition to PV generation, the demand can be met through BESS discharge or electricity supplied from the external power grid. The primary role of the BESS in this system is to mitigate the mismatch between intermittent renewable energy generation and electricity demand, thereby ensuring a reliable and continuous power supply. BESS enables the storage of excess energy generated by PV panels for later use, particularly during periods of low or no solar generation, such as nighttime or overcast days. Surplus energy may also be exported to the external grid, contributing to potential energy sales.

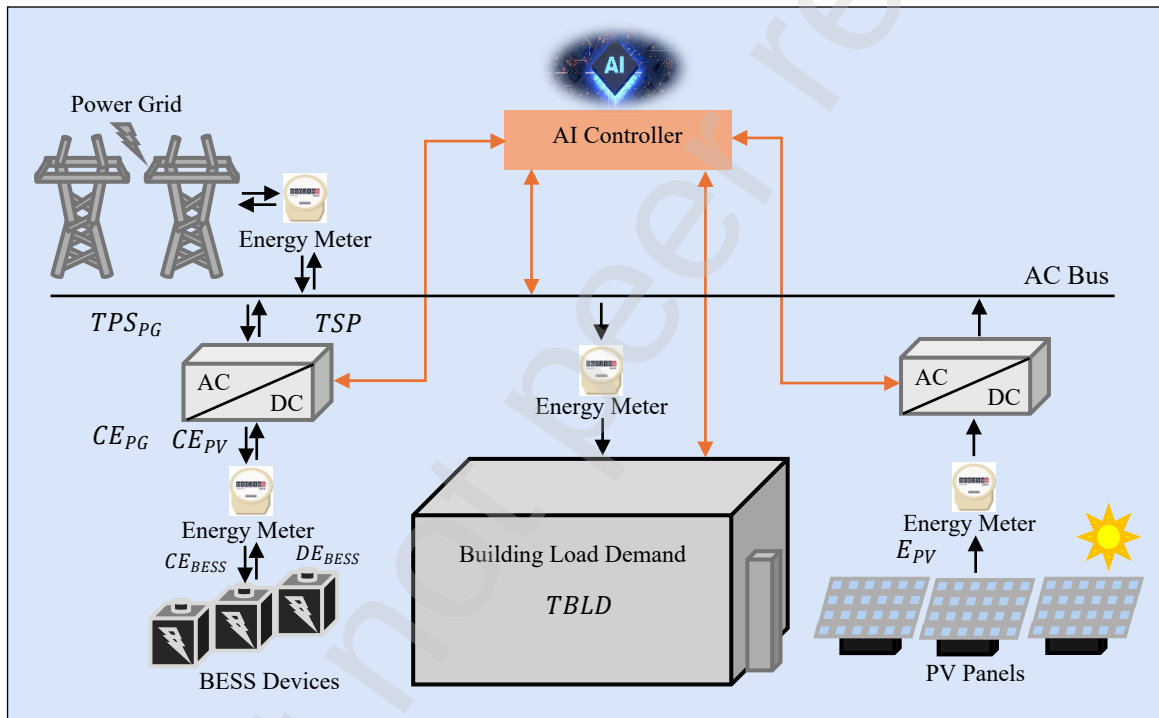


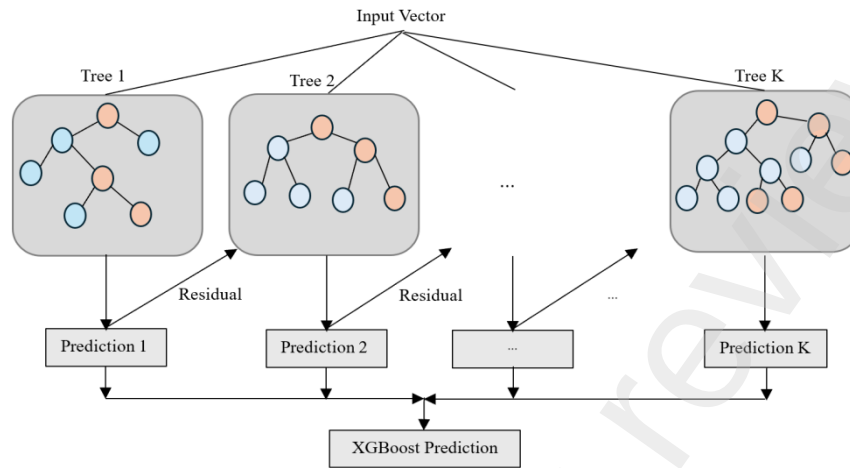
Fig. 1. Schematic of the grid-connected microgrid (MG) structure.

## 2.2. Artificial Intelligence Techniques and Performance Metrics

The literature demonstrates the widespread application of AI techniques for predicting various types of datasets [45]. In this study, two AI methods, Extreme Gradient Boosting (XGBoost) and Long Short-Term Memory (LSTM), were employed, both of which have shown strong performance, particularly with time series datasets. A brief overview of these two techniques is provided below.

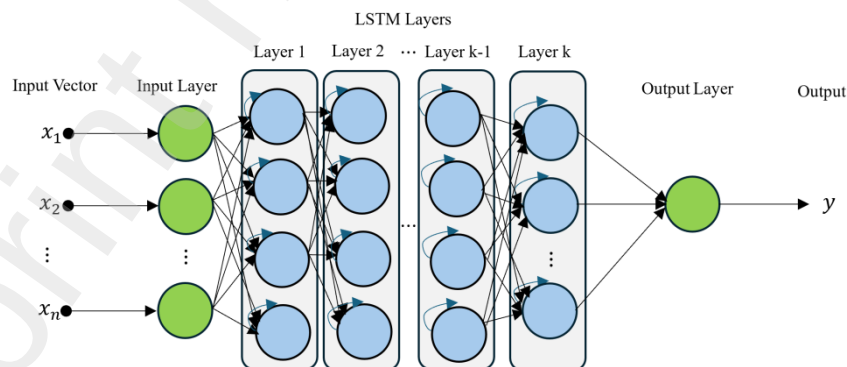
XGBoost is an efficient algorithm for constructing regression models, especially for datasets with nonlinear features. It sequentially builds a set of regression trees, starting with a simple prediction on the training data, and the amount of errors, i.e., the difference between the predicted and actual values, is calculated. Then, the first decision tree is constructed to minimize these errors, and each subsequent tree is added to correct the errors of the previous

trees. Throughout this iterative process, XGBoost minimizes a loss function that quantifies the discrepancy between predicted and actual values, ensuring improved prediction accuracy. The addition of trees continues until a predefined stopping criterion is met [45, 46]. The architecture of the XGBoost model is illustrated in Fig. 2.



**Fig. 2.** The typical architecture of Extreme Gradient Boosting (XGBoost) [45, 46].

LSTM is a type of Recurrent Neural Network (RNN) specifically designed to address the limitations of traditional RNNs in learning long-term dependencies. Unlike conventional neurons, LSTM networks utilize memory blocks connected across layers. Each LSTM block contains three non-linear gates: the input, forget, and output gates, which regulate the flow of information. The input gate determines which values should update the memory state, the forget gate decides which information to discard, and the output gate controls the information passed to the next layer based on the current input and memory state. A distinctive feature of LSTM is its separate memory cell, which allows it to retain long-term information without being influenced by short-term input fluctuations. The LSTM algorithm processes input sequences by updating the memory cell at each step [45, 47]. The architecture of the LSTM model is illustrated in Fig. 3.



**Fig. 3.** The typical architecture of the Long Short-Term Memory (LSTM) [45, 47].

Various performance metrics are available to evaluate and compare the effectiveness of AI prediction techniques. In this study, three metrics are employed to assess prediction accuracy.

**Mean Absolute Error (MAE):** This metric, calculated as the mean absolute difference between the actual and predicted values, indicates how closely the predictions align with the observed

data, regardless of the direction of the error. A lower MAE value indicates higher prediction accuracy and a better model fit [45, 47, 48]:

$$MAE = \frac{\sum_{i=1}^n |y_i - \hat{y}_i|}{n}$$

(1)

Mean Squared Error (MSE): MSE is calculated as the average of the squared differences between actual and predicted values. By squaring the errors, this metric places greater emphasis on larger deviations, making it particularly sensitive to outliers. A lower MSE value indicates a more accurate model fit [45, 47, 48]:

$$MSE = \frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}$$

(2)

Root Mean Squared Error (RMSE): This metric is the square root of the MSE. It has the same units as the original variable, making it easier to interpret than MSE. A lower RMSE indicates a better fit [45, 47, 48]:

$$RMSE = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}} \quad (3)$$

### 3. Results and Discussion

This section introduces the proposed multi-objective optimization models and the integration of AI techniques for determining the optimal size of the BESS. Additionally, it presents an AI-driven algorithm designed to manage the BESS charging and discharging schedule, enabling more efficient and adaptive energy storage operation.

#### 3.1. Proposed Multi-Objective Optimization Models

In this study, six multi-objective optimization models are proposed to determine the optimal size of BESS, including four models with two objective functions and two models with three objective functions. The following sections first present the objective functions and subsequently outline the constraints associated with the optimization models.

##### 3.1.1. Model Objective Functions

The first objective function (Minimizing the total costs): The first objective function aims to minimize the total costs, calculated as follows:

$$f_1 = Cost_{Total\_Cap} + Cost_{Total\_OM} + Cost_{Total\_ENS} + Cost_{Total\_DEC} - Income_{Total} \quad (4)$$

where  $Cost_{Total\_Cap}$  denotes the total capital cost,  $Cost_{Total\_OM}$  represents the total operation and maintenance cost,  $Cost_{Total\_ENS}$  corresponds to the economic penalty for the unsupplied energy,  $Cost_{Total\_DEC}$  is the cost of the electric energy withdrawn from the grid, and  $Income_{Total}$  is the revenue from the electricity injected into the grid.

The calculation of each term is explained below.

The total capital cost (€) is related to PV and BESS units [49]:

$$Cost_{Total\_Cap} = Cost_{Cap\_BESS} + Cost_{Cap\_PV} \quad (5)$$

The BESS capital cost (€) depends on the rated energy capacity (kWh) and rated power (kW), which is expressed as [49]:

$$Cost_{Cap\_BESS} = \sum_k [(Cost_{BESS\_kWh} * ES_{BESS\_k}^{rated}) + (Cost_{BESS\_kW} * DP_{BESS\_k}^{rated})] \quad (6)$$

The PV capital cost (€) depends on the rated power (kW):

$$Cost_{Cap\_PV} = \sum_i (Cost_{PV\_kW} * P_{PV\_i}^{rated}) \quad (7)$$

The total operation and maintenance cost (€) is calculated as follows [49]:

$$Cost_{Total\_OM} = Cost_{BESS\_OM} + Cost_{PV\_OM} \quad (8)$$

The BESS operation and maintenance cost (€), depending on the energy storage at hour t (kWh):

$$Cost_{BESS\_OM} = \sum_k \sum_t Cost_{BESS\_k\_OM\ t} \quad (9)$$

$$Cost_{BESS\_k\_OM\ t} = OMC_{BESS\_k} \times ES_{BESS\_k\ t} \quad (10)$$

The PV operation and maintenance cost (€) depends on the electric generation at hour t:

$$Cost_{PV\_OM} = \sum_i \sum_t Cost_{PV\_i\_OM\ t} \quad (11)$$

$$Cost_{PV\_i\_OM\ t} = OMC_{PV\_i} \times APG_{PV\_i\ t} \quad (12)$$

The economic penalty for the unsupplied electric energy (€) is [21]:

$$Cost_{Total\_ENS} = \sum_t Cost_{ENS\ t} \quad (13)$$

$$Cost_{ENS\ t} = [(Vul * DCC_t) + ISCO_t] * P_{ENS\ t} \quad (14)$$

The total energy cost for the electricity withdrawn from the grid corresponds to (€):

$$Cost_{Total\_DEC} = \sum_t (TPS_{PG\ t} \times UC_{Sup\ t}) = \sum_t [(TBLD_{PG\ t} + CP_{PG\ t}) \times UC_{Sup\ t}] \quad (15)$$

The total income (€) is related to the electric energy injected into the grid and is calculated as follows:

$$Income_{Total} = \sum_t (TSP_t \times UINC_t) \quad (16)$$

The income is only possible when the total energy supplied (PV generation) equals or exceeds the total demand. Therefore:

$$TSP_t = \begin{cases} 0, & \text{if } TPS_t - TLD_t \leq 0 \\ > 0, & \text{if } TPS_t - TLD_t > 0 \end{cases} \quad (17)$$

The second objective function (Minimizing the total costs, including the effect of BESS cyclic aging and PV depreciation): This objective function is incorporated into the model to control the degradation of BESS and PV performance over time by adding the costs of the BESS cyclic aging effect and the depreciation of the PV panels to the primary cost function. This objective function represents an enhanced version of  $f_1$ , which not only minimizes the costs outlined in the first objective function but also incorporates the minimization of costs related to the cyclic aging effect of the BESS and PV panel depreciation. In practical implementations of the multi-objective optimization models, one of these two objective functions is typically selected based on the specific analysis goal.

Lithium-ion batteries used in BESS have a limited cycle life, and several factors affect their aging process. Some of these factors, such as ambient temperature and cumulative operational time, are independent of the strategies employed and cannot be controlled. However, others, including the charge/discharge strategy and the depth of discharge (DOD) per cycle, are manageable and significantly impact the cyclic aging of BESS [49]. A deeper DOD tends to accelerate BESS aging and reduce cycle life, whereas shallower discharge cycles can extend BESS cycle life. Furthermore, charge and discharge rates must be carefully regulated, as high rates can elevate the BESS temperature and contribute to premature degradation [32, 49].

Table 4 presents experimental data illustrating the relationship between cycle life and different state-of-charge (SOC) levels for two types of lithium-ion batteries. Here, cycle life is defined as the number of charge/discharge cycles until the available discharging capacity or efficiency drops to 80%. The findings highlight a strong dependency of the battery degradation rate on the DOD [49].

**Table 4.** Experimental data on the relationship between cycle life and different DOD [49].

| Lithium-ion type    | DOD (%)            | DOD (%)            | DOD (%)            | DOD (%)            |
|---------------------|--------------------|--------------------|--------------------|--------------------|
|                     | Cycle life (times) | Cycle life (times) | Cycle life (times) | Cycle life (times) |
| Li/MoS <sub>2</sub> | 100%               | 80%                | 50%                | 30%                |
|                     | ~ 336              | ~ 534              | ~ 1080             | ~ 2620             |
| LiFePO <sub>4</sub> | 80%                | 50%                | 20%                | 10%                |
|                     | ~ 700              | ~ 1175             | ~ 2234             | ~ 5369             |

The throughput energy of a BESS is expressed as the total amount of energy flowing into and out of the BESS until the capacity is reduced to a level where the BESS needs to be replaced [49]. DOD optimization can lead to extending the cycle life and throughput lifetime of BESS. The BESS cycle life has an inverse exponential relationship with DOD as follows:

$$Cycle\_Life_{BESS_k} = a * DOD_{BESS_k}^{-b} * exp^{-c * DOD_{BESS_k}} \quad (18)$$

where the coefficients a, b, and c depend on the BESS technology [49]. The values for the Li/MoS<sub>2</sub> type are 25,929, 0.823, and 0.00822, respectively, and for the LiFePO<sub>4</sub> type, they are 30,650, 0.762, and 0.00888, respectively [49].

In this study, charging and discharging power constraints are incorporated into the model to prevent accelerated degradation of BESS cycle life, based on optimized charge and discharge strategies. Additionally, a recommended DOD of 80% of the total capacity is adopted for lithium-ion batteries to enhance performance [49].

To take into account the impact of BESS destruction, the destruction-related costs have been added to the original cost function ( $f_1$ ). The cost per cycle corresponding to the DOD value to consider the effect of cyclic aging on the optimal BESS size is calculated as follows:

$$Cost_{BESS\_DOD} = \sum_k \frac{(Cost_{BESS_k\_kWh} * ES_{BESS_k}^{rated}) + (Cost_{BESS_k\_kW} * DP_{BESS_k}^{rated})}{Cycle\_Life_{BESS_k}} \quad (19)$$

Additionally, PV depreciation cost is calculated as follows:

$$Cost_{DEP\_PV} = \sum_i \frac{(Cost_{PV\_t\_kW} * P_{PV\_i}^{rated})}{Serv\_Life_{PV\_i}} \quad (20)$$

The total depreciation cost is calculated as follows (€):

$$Cost_{Total\_DEP} = Cost_{BESS\_DOD} + Cost_{DEP\_PV} \quad (21)$$

Thus, the second objective function is created as follows:

$$f_2 = Cost_{Total\_DEP} + Cost_{Total\_OM} + Cost_{Total\_ENS} + Cost_{Total\_DEC} - Income_{Total} \quad (22)$$

The third objective function (Maximizing the self-sufficiency ratio): The third objective function is designed to maximize the ratio of PV production and battery discharge to total energy consumption. The objective function is described as follows [13]:

$$f_3 = \frac{\sum_t (E_{PV\_t} + DE_{BESS\_t})}{\sum_t TLD_t} \quad (23)$$

The fourth objective function (Maximizing system reliability): The fourth objective function is designed to maximize system reliability. One of the essential factors in the reliability of MG systems is the reliability index, which is related to the loss of power supply probability (LPSP). The index represents the percentage of energy supply that satisfies the load demand, indicating the reliability of the power supply for the load, and corresponds to the ratio of energy provided by the system in a specific time period (t) to the corresponding load demand in the same period [8]. The fourth objective function is defined as system reliability (dividing the total energy supply by the total load demand) as follows, which should be maximized [22]:

$$f_4 = \frac{\sum_t TES_t}{\sum_t TLD_t} \quad (24)$$

According to the objective functions defined above, in this study, six different multi-objective optimization problems (MOPs) are defined as follows:

$$\begin{aligned} \text{MOP1: } & \begin{cases} \text{Minimize } f_1 \\ \text{Maximize } f_3 \end{cases}, & \text{MOP2: } & \begin{cases} \text{Minimize } f_2 \\ \text{Maximize } f_3 \end{cases}, \\ \text{MOP3: } & \begin{cases} \text{Minimize } f_1 \\ \text{Maximize } f_4 \end{cases}, & \text{MOP4: } & \begin{cases} \text{Minimize } f_2 \\ \text{Maximize } f_4 \end{cases}, \\ \text{MOP5: } & \begin{cases} \text{Minimize } f_1 \\ \text{Maximize } f_3 \\ \text{Maximize } f_4 \end{cases}, & \text{MOP6: } & \begin{cases} \text{Minimize } f_2 \\ \text{Maximize } f_3 \\ \text{Maximize } f_4 \end{cases} \end{aligned} \quad (25)$$

According to the proposed multi-objective optimization models, a set of Pareto optimal solutions will be available, rather than just one optimal solution. This has a significant advantage: the decision maker can choose one as the optimal solution according to other management criteria.

### 3.1.2. Model Constraints

This section introduces the model constraints.

- a. Energy balance

The energy supplied by the grid is estimated based on the energy balance of the system, considering both the electricity generated locally and the load demand. The energy balance constraint is given as:

$$(TBLD_{PGt} + TBLD_{PVt} + DE_{BESS_t}) = (CE_{PGt} + CE_{PVt} + TSE_t + E_{NS_t}) \quad (26)$$

where:

$$TES_t = TES_{PGt} + E_{PVt} + DE_{BESS_t} \quad (27)$$

$$TES_{PGt} = TBLD_{PGt} + CE_{PGt} \quad (28)$$

$$E_{PVt} = TBLD_{PVt} + CE_{PVt} + TSE_{PVt} \quad (29)$$

$$CE_{BESS_t} = CE_{PGt} + CE_{PVt} \quad (30)$$

The total demand at hour t satisfies the following constraint:

$$TLD_t \leq TES_t = TES_{PGt} + E_{PVt} + DE_{BESS_t} \quad (31)$$

Considering that the capacity of the transmission line is limited, inequality (32) is defined for the grid power:

$$TES^{min} \leq TES_{PGt} \leq TES^{max} \quad (32)$$

On the other hand, the surplus power generation available for sale in the power grid must satisfy the following constraint:

$$0 \leq TSE_t \leq TES_t - TLD_t \quad (33)$$

#### b. BESS Model and Constraints

The energy capacity model of each BESS device is shown as follows [32, 40, 49]:

$$ES_{BESS_k t} = ES_{BESS_k t-1}(1 - PLR_{BESS_k}) + \xi_{BESS_k t}^{charg} \eta C_{BESS_k} CE_{BESS_k t} - \xi_{BESS_k t}^{discharg} \frac{DE_{BESS_k t}}{\eta D_{BESS_k}} \quad (34)$$

where  $\xi_{BESS_k t}^{charg}$  and  $\xi_{BESS_k t}^{discharg}$  are binary variables related to the charging and discharging status of BESS at time t:

$$\xi_{BESS_k t}^{charg} = \begin{cases} 1 & \text{if charging mode} \\ 0 & \text{if discharging or idle mode} \end{cases} \quad (35)$$

$$\xi_{BESS_k t}^{discharg} = \begin{cases} 0 & \text{if charging or idle mode} \\ 1 & \text{if discharging mode} \end{cases} \quad (36)$$

If the BESS is in the idle state, both variables are equal to zero; otherwise, only one of the charging or discharging states can occur in any given time interval. The following constraint is defined:

$$\xi_{BESS_k t}^{charg} + \xi_{BESS_k t}^{discharg} \leq 1 \quad (37)$$

On the other hand, equation (34) can be rewritten as the following linear equation:

$$ES_{BESS_k t} = ES_{BESS_k t-1}(1 - PLR_{BESS_k}) + \eta C_{BESS_k} CE_{BESS_k t} - \frac{DE_{BESS_k t}}{\eta D_{BESS_k}} \quad (38)$$

In this case, in addition to constraint (37), the following two constraints are added to the model:

$$CP_{BESS_k}^{min} t \xi_{BESS_k t}^{charg} \leq CE_{BESS_k t} \leq CP_{BESS_k}^{max} t \xi_{BESS_k t}^{charg} \quad (39)$$

$$DP_{BESS_k}^{min} t \xi_{BESS_k t}^{discharg} \leq DE_{BESS_k t} \leq DP_{BESS_k}^{max} t \xi_{BESS_k t}^{discharg} \quad (40)$$

These two constraints, along with constraints (37) and (38), limit the charging and discharging energy of the k-th BESS device and prevent them from occurring simultaneously at hour t.

The energy storage of the kth BESS device at the time t should satisfy the following constraint [11]:

$$ES_{BESS_k}^{min} \leq ES_{BESS_k t} \leq ES_{BESS_k}^{max} \quad (41)$$

Energy storage and charge/discharge power constraints are essential to the safe and efficient operation of BESS [49].

Let  $ES_{BESS_k 0}$  be the remaining energy at the beginning, and  $ES_{BESS_k n}$  be the remaining energy at the end of the day. The remaining energy at the end of the day should be the same as that at the beginning of the day, which is represented as follows [10]:

$$ES_{BESS_k 0} = ES_{BESS_k n} \quad (42)$$

The two main factors that affect the lifetime of a BESS are the number of cycles and the state of charge (SOC) [9]. The SOC of the kth BESS device at the time t can be given as follows [11, 13]:

$$SOC_{BESS_k t} = \frac{ES_{BESS_k t}}{ES_{BESS_k}^{rated}} * 100\% \quad (43)$$

SOC should be controlled within a proper range [11, 49]:

$$SOC_{BESS_k}^{min} \leq SOC_{BESS_k t} \leq SOC_{BESS_k}^{max} \quad (44)$$

The following constraint establishes a relationship between minimum energy capacity, rated energy capacity, and depth of discharge [22, 31]:

$$ES_{BESS_k}^{min} = (1 - DOD_{BESS_k}) * ES_{BESS_k}^{rated} \quad (45)$$

As shown in equation (46), the total capital cost of the BESS cannot exceed the budget limits:

$$Cost_{Cap\_BESS} \leq Budget_{BESS} \quad (46)$$

Tactically, BESS discharge only makes economic sense when there is a lack of power or the high cost of power from the grid. Therefore, the energy generated by BESS discharge should never be sold as surplus power to the grid. The following constraints are added to the model to prevent the simultaneous discharge of BESS and sale of power to the grid:



$$\beta_t^{TPS} + \xi_{BESS_k t}^{discharg} \leq 1 \quad (47)$$

$$0 \leq TSE_t \leq \beta_t^{TPS} P_{PV}^{rated} t \quad (48)$$

where  $\beta_t^{TPS}$  is a binary variable related to the selling surplus generation to the grid over the period  $t$ :

$$\beta_t^{TPS} = \begin{cases} 1 & \text{if selling surplus mode} \\ 0 & \text{if else} \end{cases} \quad (49)$$

### c. Photovoltaic Panel Model and Constraints

The photovoltaic (PV) energy generation depends on the external solar radiation intensity, and the installation area and energy conversion efficiency of the installed PV panel, formulated as follows [13, 27]:

$$E_{PV_i t} = SRI_{PV_i t} * IA_{PV_i} * \eta_{PV_i t} \quad (50)$$

The energy conversion efficiency of the panel varies with the external solar radiation intensity and outdoor ambient temperature, which can be calculated as [13, 27]:

$$\eta_{PV_i t} = a_1 \left[ \left( \frac{SRI_{PV_i t}}{SRI_0} \right)^{a_2} + a_3 \left( \frac{SRI_{PV_i t}}{SRI_0} \right) \right] \left[ 1 + \left( \frac{T_{out t}}{T_0} \right)^{a_2} + a_5 \left( \frac{AM_t}{AM_0} \right) \right] \quad (51)$$

where  $a_1$ ,  $a_2$ ,  $a_3$ ,  $a_4$ , and  $a_5$  are constant coefficients, which are 0.28204, 0.39668,  $-0.44730$ ,  $-0.092864$ , and 0.016010, respectively. The PV output power constraint is as follows:

$$0 \leq E_{PV_i t} \leq P_{PV_i}^{rated} t \quad (52)$$

This study utilizes AI techniques to predict and estimate the hourly power generated by PV panels as accurately as possible.

## 3.2. AI-Based Methods for Optimal Sizing and Managing BESS Charge and Discharge Schedule

This section introduces a method that leverages AI techniques to support the determination of the optimal size of the BESS. Subsequently, an AI-based algorithm is presented for managing BESS charging and discharging schedules, enabling adaptive and data-driven control of energy storage operations.

### 3.2.1. AI-Based Methods for Optimal Sizing of BESS

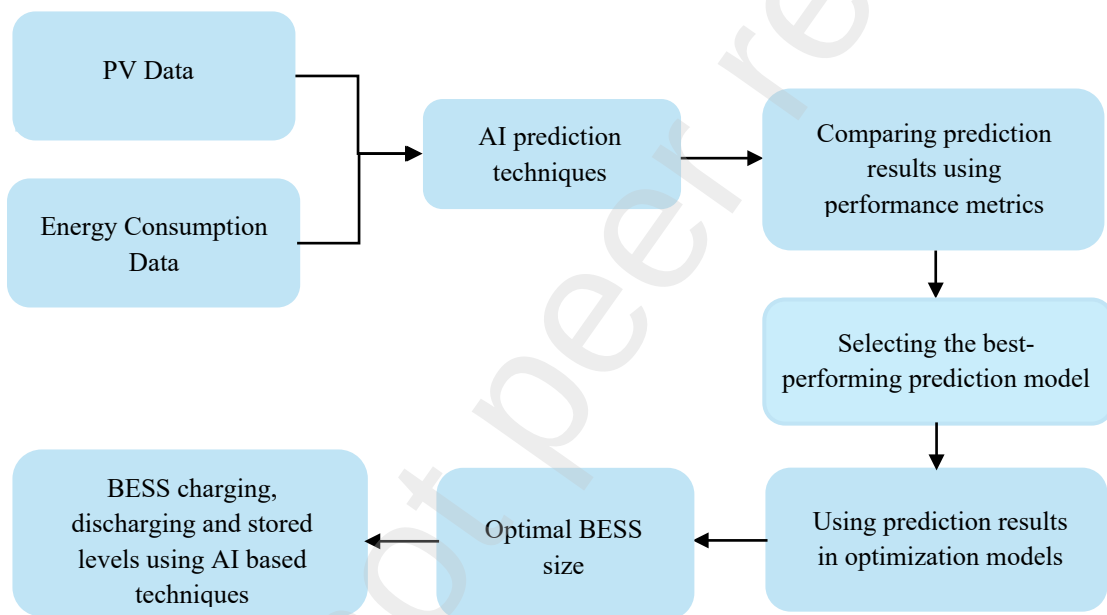
In many real-world applications, the optimal sizing of a BESS is determined based on historical system data, with the additional objective of ensuring suitability for future operational conditions. However, accurately predicting energy consumption and PV generation remains a significant challenge due to the inherent variability and intermittency of these sources. Precise prediction of both energy demand and PV output is therefore crucial for optimizing BESS sizing decisions.

Fig. 4 illustrates the proposed framework for integrating AI techniques to optimize the sizing of the BESS and to manage its charging and discharging operations. In the first stage, AI

models are employed to forecast PV generation and energy consumption. The model with the lowest prediction error is selected to provide these forecasts, which serve as critical inputs to the multi-objective optimization models. In the second stage, after determining the optimal BESS size through these optimization models, AI techniques are further utilized to schedule charging and discharging operations and to regulate the targeted energy storage levels throughout the day.

### 3.2.1. AI-Based Algorithm for Managing BESS Charge and Discharge Schedule

The service life of lithium-ion batteries in BESS is influenced by several key factors, including ambient temperature, the number of charge-discharge cycles, and DOD. Additionally, excessively high charging and discharging rates can lead to elevated internal temperatures, which accelerate battery degradation and shorten the operational lifespan [32].



**Fig. 4.** The comprehensive framework for leveraging AI techniques in the optimal sizing of BESS, as well as in the scheduling of charging, discharging, and energy storage level management.

To address these challenges, this section presents an AI-based algorithm designed to intelligently schedule the charging and discharging operations of a BESS within a grid-connected microgrid. The algorithm leverages AI models, specifically XGBoost and LSTM, to forecast PV generation and energy consumption over a 24-hour horizon. These predictions inform the BESS scheduling strategy, which aims to enhance PV self-consumption, increase system reliability, and extend battery life. The algorithm incorporates the following key components:

- Selection of the most accurate AI forecasting model based on RMSE evaluation.
- Implementation of a dynamic target state-of-charge (SoC) curve.
- Hourly scoring logic for prioritizing charge/discharge decisions.
- Event-driven triggers to adapt to unexpected changes in system conditions.

This approach enables more responsive and efficient energy management, supporting both technical performance and long-term battery health.

The workflow of the proposed AI-based algorithm for Battery Energy Storage System (BESS) scheduling is outlined as follows:

Step 1: Input Parameters:

- BESS specifications: capacity and power rating.
- Forecasted photovoltaic (PV) generation and load demand (via AI models).
- Electricity purchase and export tariffs.
- Initial and target state of charge (SoC) levels.

Step 2: Generation of Target SoC Curve:

- Calculate hourly net demand:  $\text{Net Demand}(t) = E_{PV_t}^{\text{Forecast}} - TLD_t^{\text{Forecast}}$  (which  $E_{PV_t}^{\text{Forecast}}$  and  $TLD_t^{\text{Forecast}}$  are the forecasted PV electric generation and load demand at time  $t$ , respectively).
- Normalize and invert the net demand profile to construct a desired SoC trajectory.
- Adjust initial and final SoC values to ensure operational readiness and maintain battery stability.

Step 3: Priority Scoring Mechanism:

- For each hour, compute a composite score to prioritize energy actions:  
$$\text{Score}(t) = w_1 \times \text{PV}(t) - w_2 \times \text{Load}(t) - w_3 \times \text{Price}(t).$$
- Normalize scores to a range of  $[-1, 1]$  for consistent decision-making.

Step 4: Event-Based Triggering Logic:

- If  $\text{Score}(t) \geq \text{threshold}$  (e.g., +DOD): Charging is permitted.
- If  $\text{Score}(t) \leq \text{threshold}$  (e.g., -DOD): Discharging is permitted.
- Charging is prioritized using surplus PV; discharging is targeted to offset load.

Step 5: Logic decision for BESS Actions:

- Charge if:  $\text{Score} \geq 0$  and  $\text{SoC} < \text{Maximum}$ .
- Discharge if:  $\text{Score} < 0$  and  $\text{SoC} > \text{Minimum}$ .
- Charging/discharging adheres to SoC boundaries and BESS constraints.
- PV surplus is used first for charging; the grid is used only when necessary.

Step 6: Output Generation:

- Hourly BESS charging power profile.
- Hourly BESS discharging power profile.
- Resulting hourly energy stored in the BESS.

This algorithm enables adaptive, forecast-driven energy scheduling that balances efficiency, reliability, and battery health in real-time microgrid operation.

#### 4. Computational Experiments

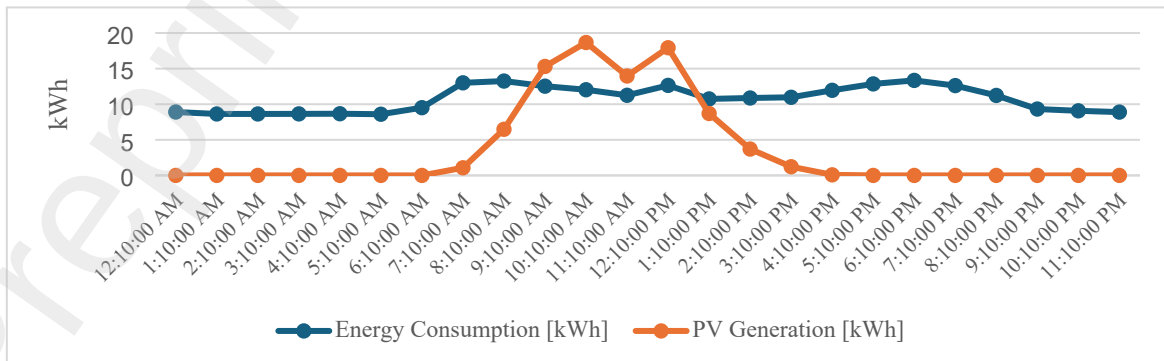
This section presents an evaluation of the proposed multi-objective optimization models and the integrated framework that leverages AI techniques for optimal BESS sizing and for scheduling charging, discharging, and managing energy storage levels. The evaluation is carried out through a series of computational experiments.

Data analysis was performed using Python coding within the Jupyter extension of Visual Studio Code. A trial-and-error approach was employed to identify the best-performing prediction technique. To solve the multi-objective optimization problems, the “Global Solve” feature of the global optimization package in Maple version 18.01 was utilized. Table 5 summarizes the main characteristics of the BESS and power grid utilized in the following case studies.

**Table 5.** Main characteristics of the BESS and power grid [3].

| Parameters                                       | Values      | Parameters                                                                      | Values     |
|--------------------------------------------------|-------------|---------------------------------------------------------------------------------|------------|
| Capacity rating cost ( $Cost_{BESS\_kWh}$ )      | 250 €/kWh   | Power rating cost ( $Cost_{BESS\_kW}$ )                                         | 250 €/kW   |
| Operation and maintenance cost ( $OMC_{BESS}$ )  | 0.005 €/kWh | Lower value of SOC ( $SOC_{BESS}^{min}$ )                                       | 15%        |
| Charging efficiency ( $\eta C_{BESS}$ )          | 0.95        | Upper value of SOC ( $SOC_{BESS}^{max}$ )                                       | 90%        |
| Discharging efficiency ( $\eta D_{BESS}$ )       | 0.98        | Min DOD                                                                         | 20%        |
| Self-discharge rate ( $PLR_{BESS}$ )             | 0.0002      | Maximum cycle number                                                            | 5000       |
| Round trip efficiency ( $\eta RT_{BESS}$ )       | 98%         | Maximum life span                                                               | 10 years   |
| Minimum energy capacity ( $ES_{BESS}^{min}$ )    | 0 kWh       | Unit cost of purchasing electric energy in intermediate period ( $UC_{Sup t}$ ) | 0.20 €/kWh |
| Lower value of charging ( $CE_{BESS}^{min}$ )    | 0 kWh       | Unit cost of purchasing electric energy in peak period ( $UC_{Sup t}$ )         | 0.35 €/kWh |
| Lower value of discharging ( $DE_{BESS}^{min}$ ) | 0 kWh       | Unit cost of purchasing electric energy off-peak period ( $UC_{Sup t}$ )        | 0.10 €/kWh |
| Unit income of electric energy ( $UINC_t$ )      |             |                                                                                 | 0.15 €/kWh |

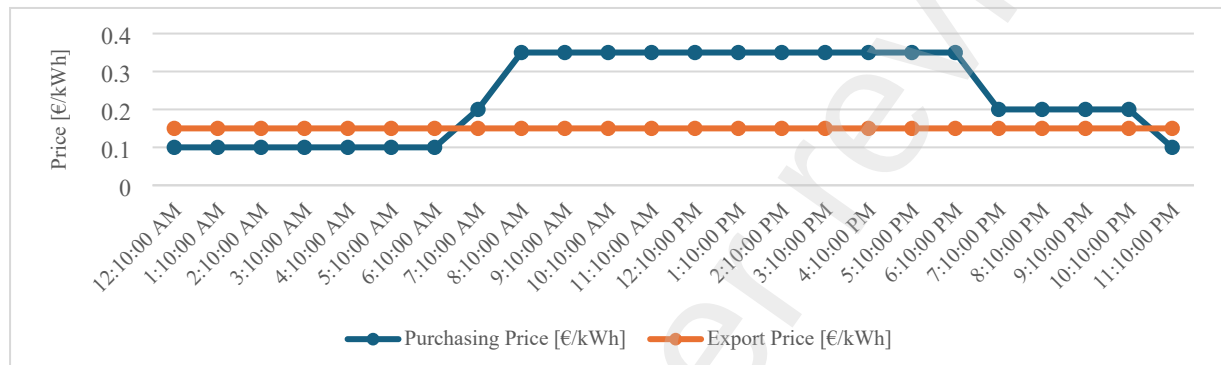
**Case Study 1.** This case study aims to demonstrate the effectiveness of the proposed multi-objective optimization models in determining the optimal sizing of BESS. The analyzed system consists of PV modules with a total nominal capacity of 50 kW, serving as the primary renewable energy source. Fig. 5 illustrates the hourly energy consumption and corresponding PV generation for a representative day. PV generation was modeled using Photovoltaic Geographical Information System (PVGIS) data for a rooftop system with south-facing panels at a 30° slope [50]. The energy consumption data used in this study represents a typical daily profile for an industrial facility with continuous operations.



**Fig. 5.** Energy consumption and total energy production from PV panels in a single day.

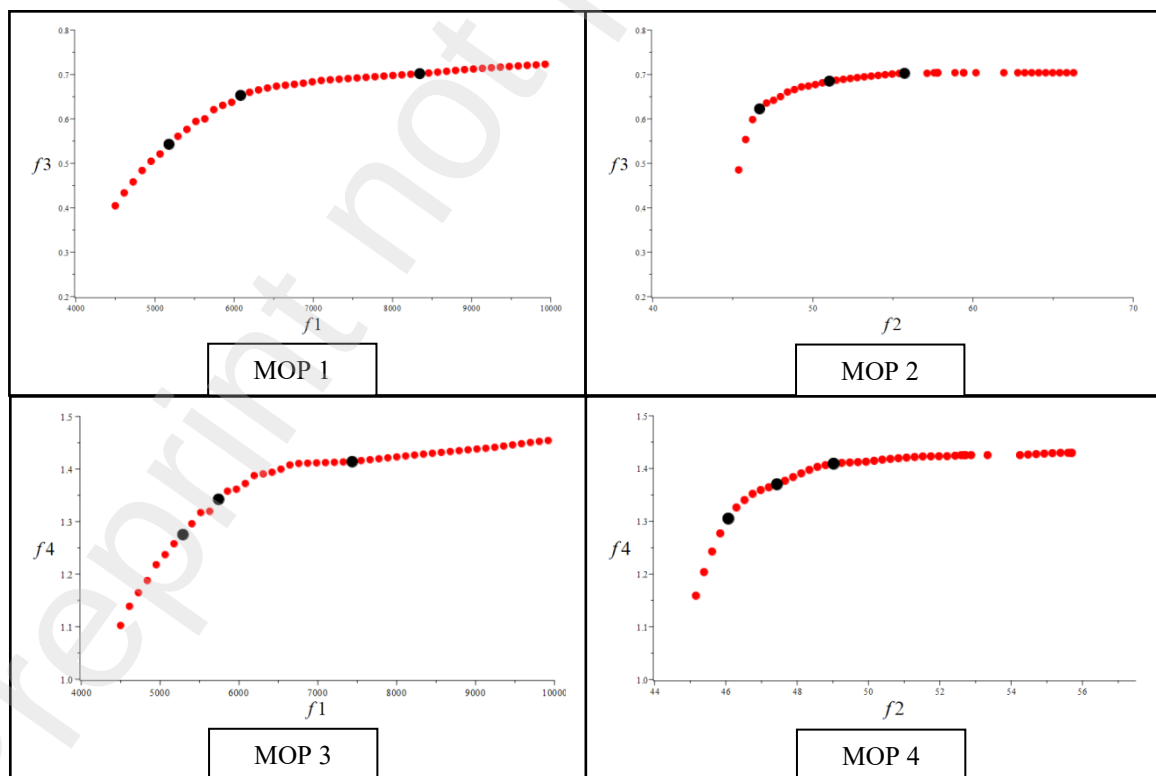
As illustrated in Fig. 5, PV energy production reaches its peak around midday, whereas peak energy consumption occurs at approximately 8:00 AM and 6:00 PM. Notably, between approximately 9:00 AM and 1:00 PM, the energy generated by the PV panels exceeds the amount of energy consumed. The presence of excess energy during certain periods suggests that incorporating a BESS would be beneficial for storing surplus PV generation. The optimal size of the BESS is determined using the proposed multi-objective optimization models.

In the analyzed system, the export tariff for supplying energy to the external grid is 0.15 €/kWh. The electricity purchase price varies throughout the day, with rates of 0.35 €/kWh during peak hours, 0.10 €/kWh during off-peak hours, and 0.20 €/kWh during intermediate hours. Fig. 6 illustrates the time-of-day pricing structure for both electricity purchase and export.



**Fig. 6.** Electricity purchase and export prices during the day [37].

Figure 7 presents the approximate Pareto frontiers derived from the bi-objective optimization models applied in Case Study 1.



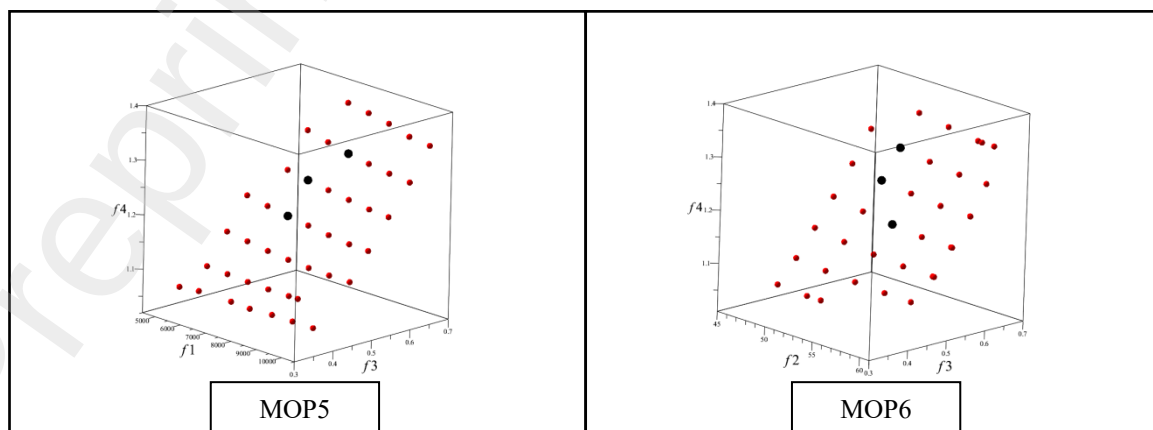
**Fig. 7.** The approximation of Pareto frontiers of bi-objective models in Case Study 1.

As illustrated in Fig. 7, the results of MOP1 indicate that the self-sufficiency ratio increases to approximately 72%. In contrast, for MOP2, where DOD is controlled, the rate rises to approximately 70%. In MOP1, integrating BESS into the system results in a steep initial increase in the self-sufficiency ratio, reaching approximately 65%. Beyond this point, further increases in BESS capacity primarily result in higher costs, with minimal improvement in the consumption ratio. A similar trend is observed in MOP2, where the ratio increases rapidly up to about 62% with the addition of BESS, after which the marginal benefit of increasing BESS capacity on the renewable energy consumption ratio diminishes. Furthermore, the results of MOP 3 and MOP 4 exhibit similar trends in the relationship between BESS sizing and system reliability. Three representative Pareto-optimal solutions were selected for further analysis from MOP 1 to MOP 4. These solutions are indicated by black circles in Fig. 7, and their corresponding details are provided in Table 6.

**Table 6.** Main characteristics of selected Pareto-optimal solutions for MOP 1- MOP 4.

| MOP   | Number | BESS           |            | Self-sufficiency ratio (%) |
|-------|--------|----------------|------------|----------------------------|
|       |        | Capacity (kWh) | Power (kW) |                            |
| MOP 1 | 1      | 16.24          | 4.02       | 54.27%                     |
|       | 2      | 16.43          | 7.64       | 65.28%                     |
|       | 3      | 23.77          | 9.45       | 70.18%                     |
| MOP 2 | 1      | 20.68          | 7.45       | 62.24%                     |
|       | 2      | 23.71          | 9.33       | 68.50%                     |
|       | 3      | 30.00          | 10.00      | 70.25%                     |
| MOP   | Number | BESS           |            | System reliability (-)     |
|       |        | Capacity (kWh) | Power (kW) |                            |
| MOP 3 | 1      | 16.40          | 4.47       | 1.28                       |
|       | 2      | 16.54          | 6.27       | 1.34                       |
|       | 3      | 19.61          | 9.99       | 1.41                       |
| MOP 4 | 1      | 20.67          | 6.47       | 1.31                       |
|       | 2      | 20.68          | 9.05       | 1.37                       |
|       | 3      | 20.72          | 10.00      | 1.41                       |

A comparison of the Pareto-optimal solutions presented in Table 6 reveals that imposing a DOD constraint to account for cyclic aging in MOP2 and MOP4 necessitates a larger BESS capacity to achieve comparable levels of self-consumption and system reliability as observed in MOP1 and MOP3, respectively. Figure 8 illustrates the approximated Pareto frontiers generated by three objective optimization models.



**Fig. 8.** The approximation of Pareto frontiers of three objective models in Case Study 1.

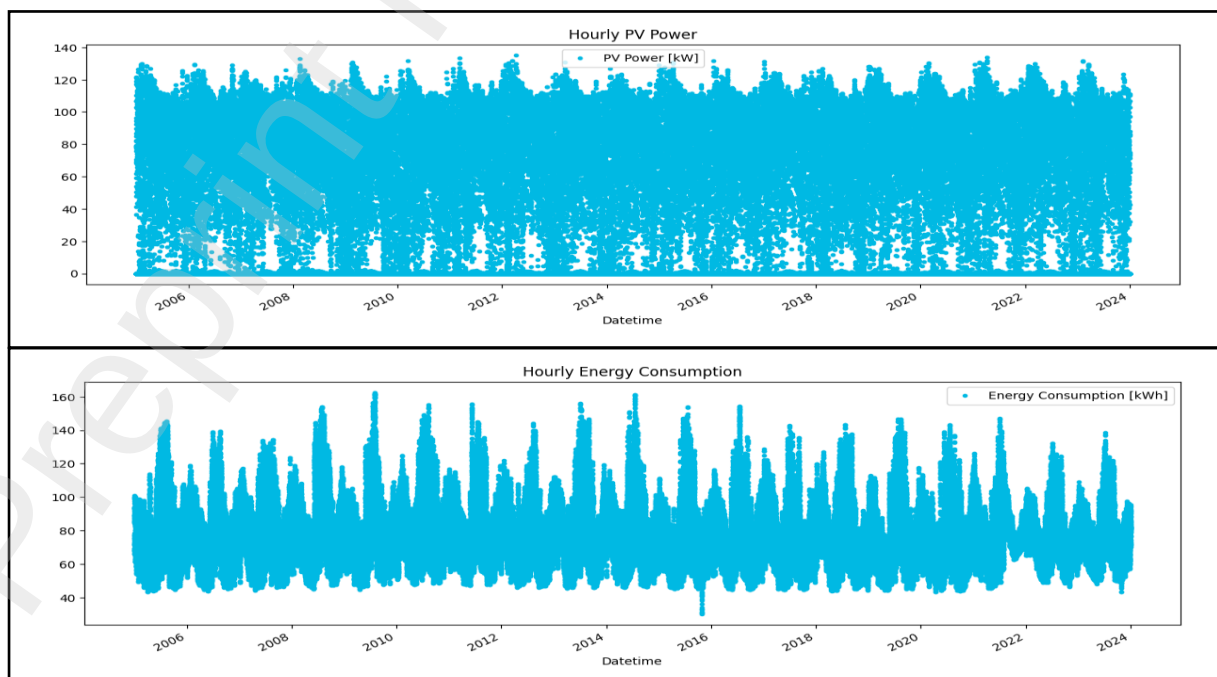
Three Pareto-optimal solutions, marked by black circles in Fig. 8, were selected for further analysis. The detailed characteristics of these solutions are provided in Table 7. The comparison of Pareto solutions from the three-objective optimization models, as presented in Table 7, confirms that imposing constraints on DOD necessitates a larger BESS capacity to attain similar levels of self-sufficiency and system reliability.

**Table 7.** Main characteristics of selected Pareto-optimal solutions for MOP 5 and MOP 6.

| MOP   | Number | BESS           |            | Self-sufficiency ratio (%) | System reliability (-) |
|-------|--------|----------------|------------|----------------------------|------------------------|
|       |        | Capacity (kWh) | Power (kW) |                            |                        |
| MOP 5 | 1      | 21.11          | 5.97       | 52.10%                     | 1.19                   |
|       | 2      | 20.93          | 6.15       | 57.15%                     | 1.25                   |
|       | 3      | 22.85          | 7.46       | 62.20%                     | 1.30                   |
| MOP 6 | 1      | 23.56          | 6.47       | 51.16%                     | 1.24                   |
|       | 2      | 27.51          | 6.50       | 51.83%                     | 1.18                   |
|       | 3      | 23.25          | 8.99       | 61.11%                     | 1.30                   |

**Case Study 2.** This case study aims to demonstrate the effectiveness of integrating AI techniques with the proposed multi-objective optimization models for determining the optimal size of BESS. The system under investigation consists of several PV modules with a total maximum power capacity of 150 kW. The energy output from PV panels varies with weather conditions, and energy consumption fluctuates due to both operational and environmental factors. The solar energy dataset utilized in this study records the power output of PV systems in kilowatts (kW), providing a measure of their performance. The energy consumption dataset also captures hourly electricity consumption, reported in kilowatt-hours (kWh).

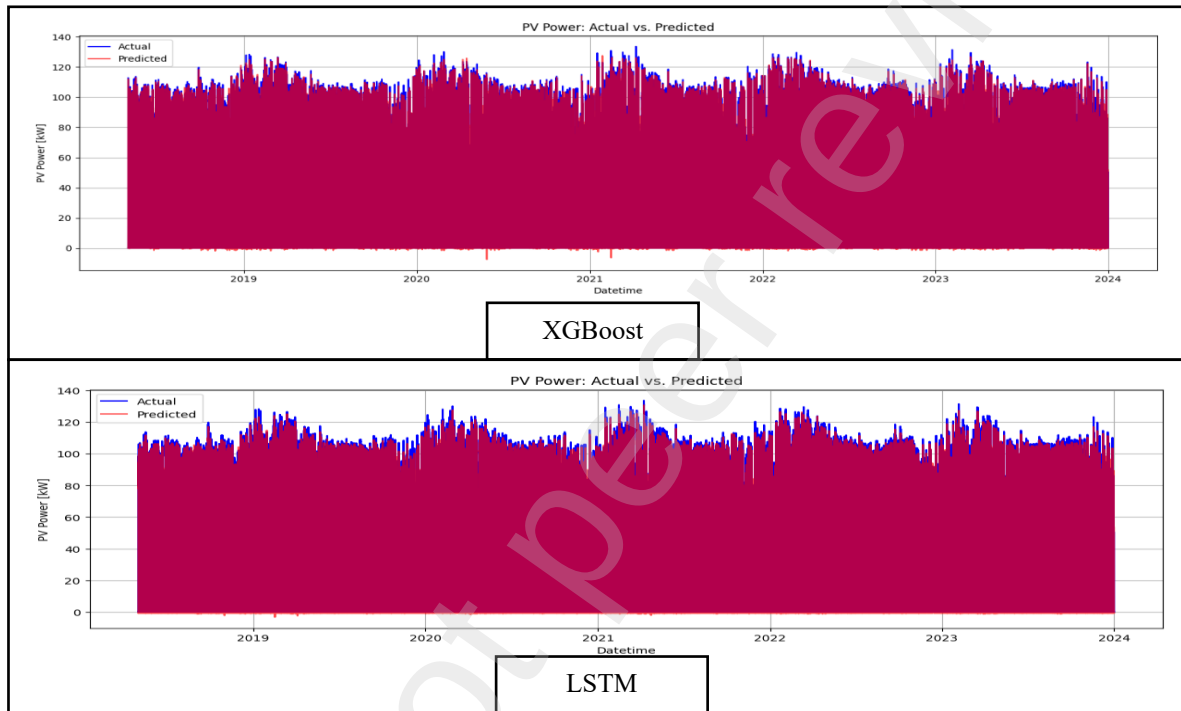
The PV dataset originates from a site located in southern Italy. It includes two components: Date and time, formatted as month/day/year/hour, and PV power (kW) [50]. The energy consumption dataset also consists of date and time, and energy consumption (kWh) [51]. Fig. 9 presents the set of reported hourly data on energy generation from PV panels and hourly energy consumption for the period spanning 2005 to 2023.



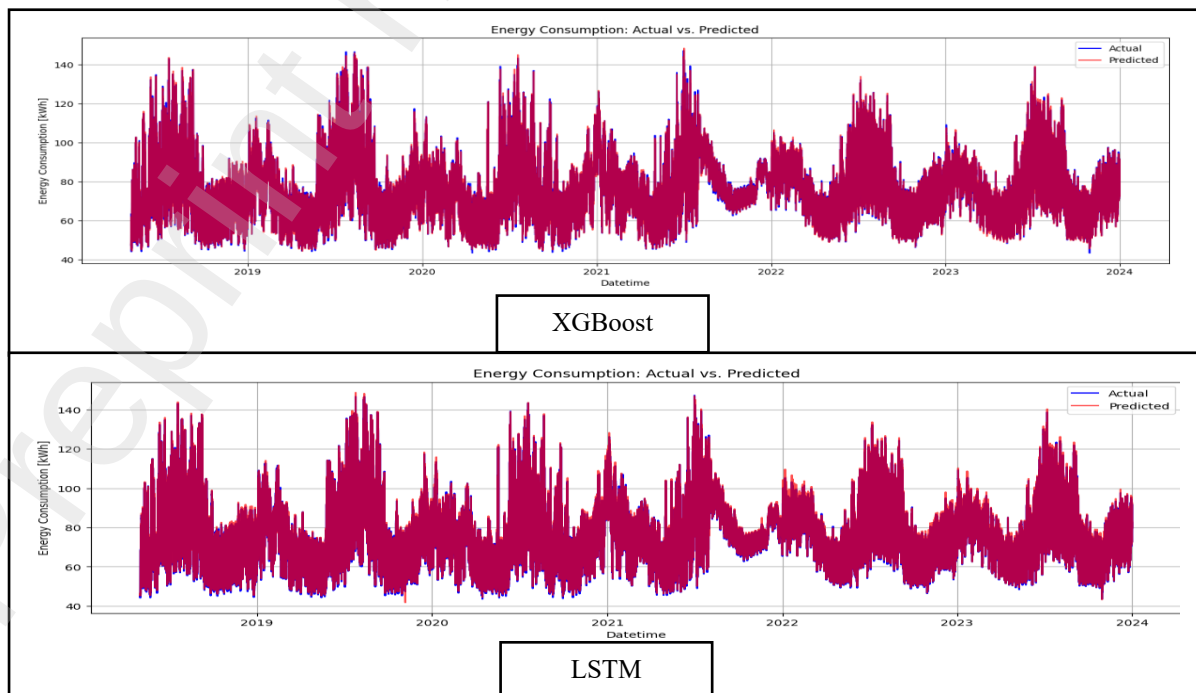


**Fig. 9.** Hourly reported PV power generation data and energy consumption from 2005 to 2023.

In both datasets, 70% of the data was used for training, and the remaining 30% was reserved for testing. The AI regression techniques XGBoost and LSTM were employed to perform predictions on the datasets. Figures 10 and 11 illustrate the prediction results of these techniques on the testing datasets for PV power generation and energy consumption, respectively. The blue segments represent the actual data, while the red segments denote the predicted values.



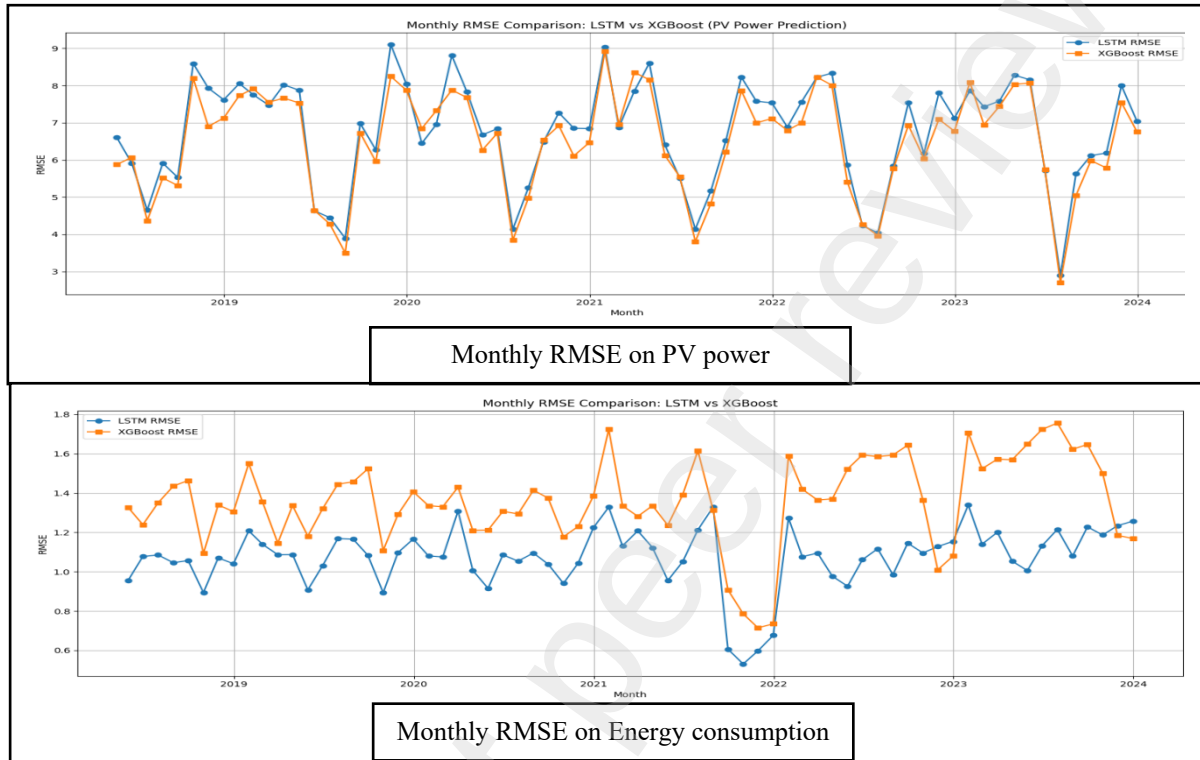
**Fig. 10.** Comparison of actual (blue) and predicted (red) data for PV power generation dataset.





**Fig. 11.** Comparison of actual (blue) and predicted (red) data for the energy consumption dataset.

Initially, the most appropriate prediction technique for each dataset was identified using the performance metrics described in Section 2.2. Fig. 12 compares XGBoost and LSTM based on monthly RMSE values for the PV power generation and energy consumption datasets.



**Fig. 12.** Comparison of XGboost and LSTM based on monthly RMSE values for the PV power generation and energy consumption datasets.

As shown in Fig. 12, the XGBoost technique demonstrates superior prediction performance for the PV power generation dataset, while the LSTM technique performs better for the energy consumption dataset. Table 8 presents the values of three performance metrics for these datasets.

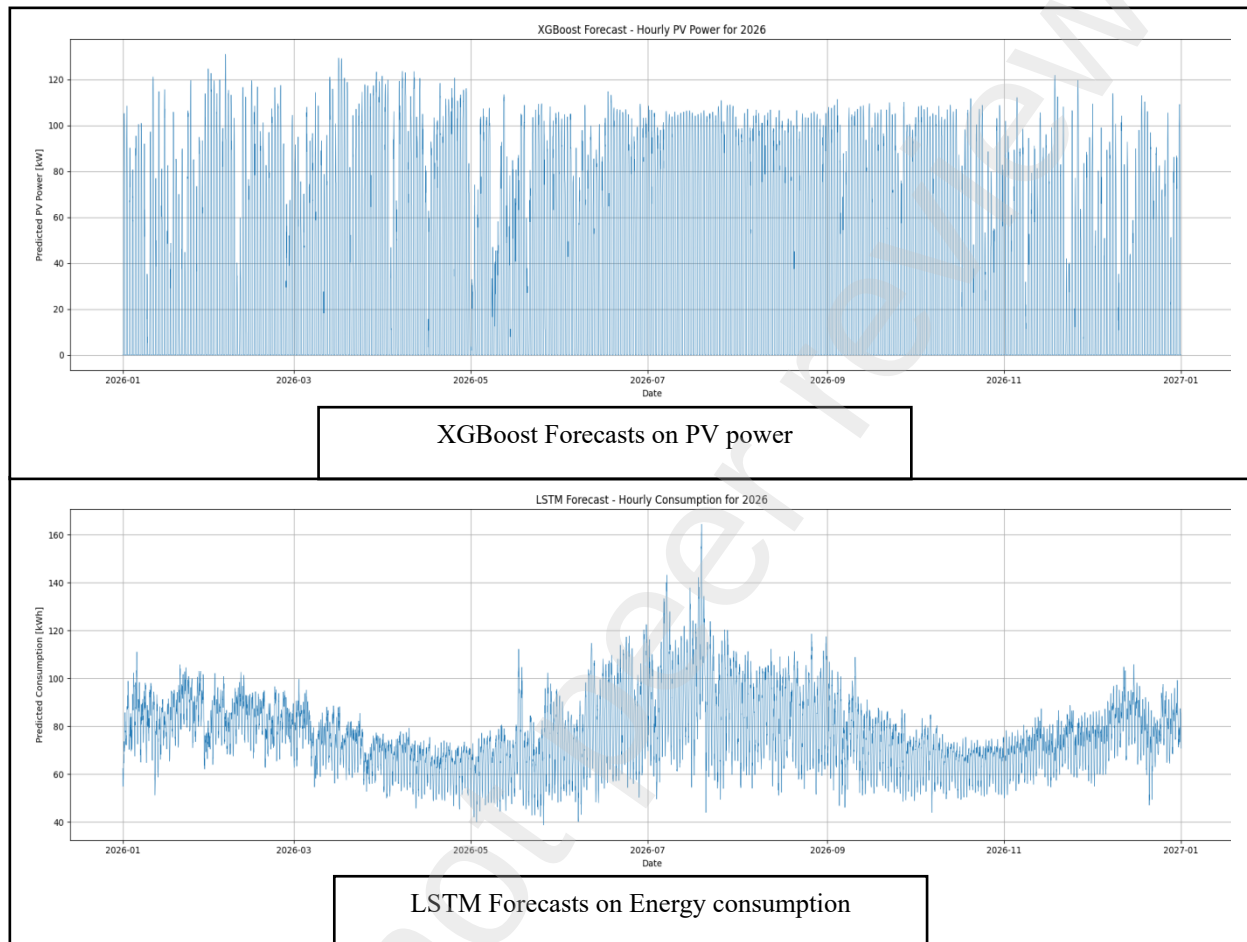
**Table 8.** The performance metrics results for AI techniques on the PV power generation and energy consumption datasets.

| Data set            | AI Model | Performance Metrics |         |        |
|---------------------|----------|---------------------|---------|--------|
|                     |          | MAE                 | MSE     | RMSE   |
| PV power generation | XGBoost  | 3.0796              | 44.0148 | 6.6344 |
|                     | LSTM     | 3.5033              | 47.5633 | 6.8966 |
| Energy consumption  | XGBoost  | 1.0344              | 1.9013  | 1.3789 |
|                     | LSTM     | 0.7807              | 1.1293  | 1.0627 |

Lower values of MAE, MSE, and RMSE indicate a better fit between the predicted and actual data. As shown in Table 8, the XGBoost technique achieves superior performance for predicting the PV power generation dataset, whereas the LSTM technique performs better for the energy consumption dataset. Accordingly, the XGBoost technique is employed for

prediction on the PV power generation dataset, while the LSTM technique is utilized for prediction on the energy consumption dataset.

Figure 13 presents the hourly forecasts of PV power generation and energy consumption for 2026, obtained using the XGBoost and LSTM techniques, respectively.



**Fig. 13.** Hourly forecasts of PV power generation using XGBoost and energy consumption using LSTM for 2026.

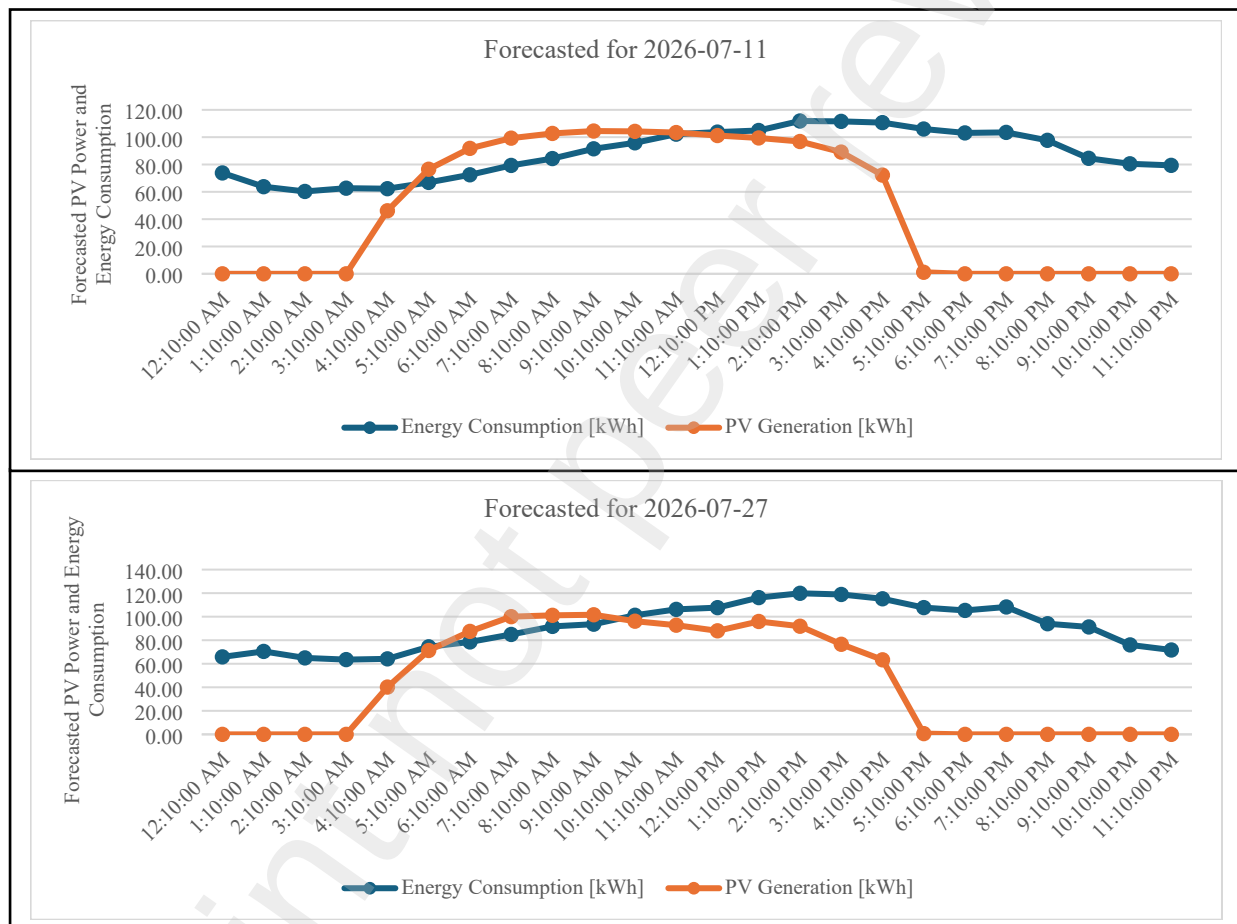
The next goal is to identify the days with the lowest prediction error to support the determination of the optimal BESS size using the proposed multi-objective optimization models. To achieve this, the ten days with the lowest prediction errors (based on RMSE values) were selected based on the XGBoost predictions for the PV power dataset and the LSTM predictions for the energy consumption dataset. The results are summarized in Table 9.

**Table 9.** The top ten days with the lowest prediction error.

| PV Power Dataset |            |          | Energy Consumption Dataset |            |          |
|------------------|------------|----------|----------------------------|------------|----------|
| Rank             | Date       | RMSE     | Rank                       | Date       | RMSE     |
| 1                | 2023-07-11 | 0.612965 | 1                          | 2019-07-27 | 0.368107 |
| 2                | 2023-07-17 | 0.633746 | 2                          | 2021-07-24 | 0.391520 |
| 3                | 2020-06-30 | 0.635217 | 3                          | 2018-10-27 | 0.407286 |
| 4                | 2023-07-23 | 0.652518 | 4                          | 2019-08-01 | 0.422893 |
| 5                | 2023-07-30 | 0.657132 | 5                          | 2020-07-05 | 0.433158 |
| 6                | 2020-07-31 | 0.681673 | 6                          | 2018-05-04 | 0.433798 |
| 7                | 2018-07-13 | 0.685598 | 7                          | 2019-02-08 | 0.437664 |

|    |            |          |    |            |          |
|----|------------|----------|----|------------|----------|
| 8  | 2023-07-16 | 0.689808 | 8  | 2020-08-11 | 0.442446 |
| 9  | 2023-06-22 | 0.691057 | 9  | 2019-07-03 | 0.444965 |
| 10 | 2022-06-02 | 0.692492 | 10 | 2019-05-27 | 0.445659 |

As shown in Table 9, the most accurate predictions for the PV power generation dataset occurred in July, with seven out of the ten days exhibiting the lowest RMSE values during this month. Similarly, July had the highest concentration of accurate predictions in the energy consumption dataset, with four of the ten days showing the lowest prediction errors. From these days, the 11th and 27th of July were selected for further analysis. Using the XGBoost technique, hourly forecasts of PV power generation were generated, while the LSTM technique was employed for hourly forecasts of energy consumption on these two selected days in July 2026. The resulting hourly predictions are illustrated in Fig. 14.

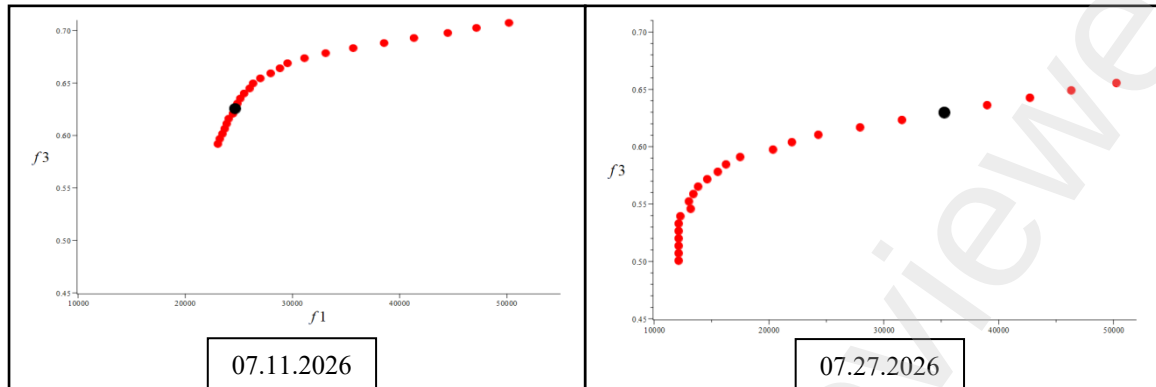


**Fig. 14.** Hourly forecasts of PV power generation and energy consumption for July 11, 2026, and July 27, 2026.

Based on the forecasted data for 2026, the results of each proposed multi-objective optimization model are analyzed for these two specific days. Fig. 15 presents the approximate Pareto frontier for MOP1.

The comparison of the MOP1 Pareto frontiers in Fig. 15 reveals that, on July 11, the self-sufficiency ratio begins at approximately 60% and increases to a maximum of around 70%. In contrast, on July 27, it starts at about 50% and rises to a maximum of approximately 65%. A Pareto-optimal solution from each day, corresponding to a self-sufficiency ratio of

approximately 63%, was selected for further analysis. The associated results are provided in Table 10.



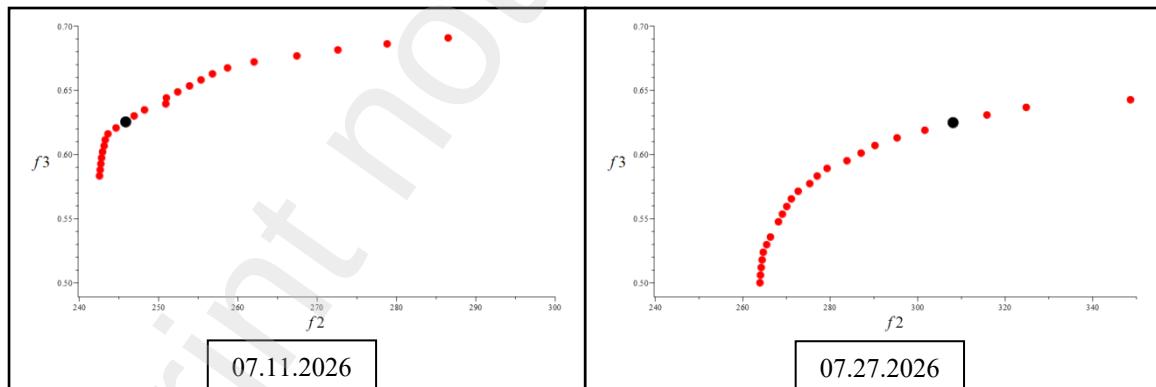
**Fig. 15.** The approximation of Pareto frontiers of MOP1 on two selected days in Case Study 2.

**Table 10.** Main characteristics of the selected Pareto-optimal solution of MOP 1 for Case Study 2.

| Date       | BESS           |            | Self-sufficiency ratio (%) |
|------------|----------------|------------|----------------------------|
|            | Capacity (kWh) | Power (kW) |                            |
| 07.11.2026 | 85.31          | 12.47      | 62.56%                     |
| 07.27.2026 | 90.12          | 48.69      | 62.97%                     |

As presented in Table 10, achieving a comparable self-sufficiency ratio on day 27 requires a BESS with greater energy capacity and power compared to day 11.

Fig. 16 illustrates the approximated Pareto frontier for MOP 2 across the two selected days, while Table 11 presents the details of the chosen Pareto-optimal solution for each day.



**Fig. 16.** The approximation of Pareto frontiers of MOP 2 on two selected days in Case Study 2.

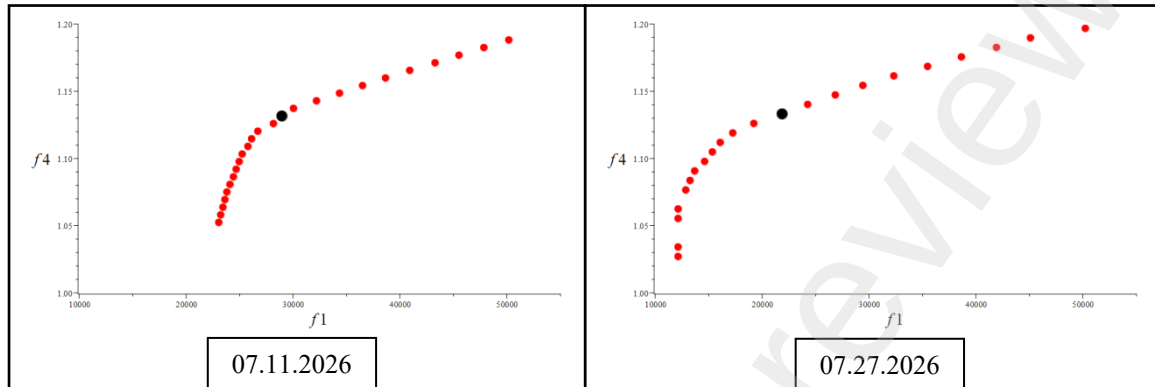
As illustrated in Fig. 16, for MOP2, the cost associated with achieving a self-sufficiency ratio of approximately 63% is lower on July 11 compared to July 27. Additionally, as shown in Table 11, the BESS capacity required on July 27 is slightly larger than that determined for July 11, while the required power for July 27 is approximately 2.5 times greater.

**Table 11.** Main characteristics of the selected Pareto-optimal solution of MOP 2 for Case Study 2.

| Date       | BESS           |            | Self-sufficiency ratio (%) |
|------------|----------------|------------|----------------------------|
|            | Capacity (kWh) | Power (kW) |                            |
| 07.11.2026 | 85.31          | 12.47      | 62.56%                     |
| 07.27.2026 | 90.12          | 48.69      | 62.97%                     |

|            |        |       |        |
|------------|--------|-------|--------|
| 07.11.2026 | 106.61 | 19.16 | 62.53% |
| 07.27.2026 | 112.43 | 50.00 | 62.48% |

Fig. 17 presents the approximated Pareto frontiers for MOP3, while Table 12 provides the details of the two Pareto-optimal solutions selected.



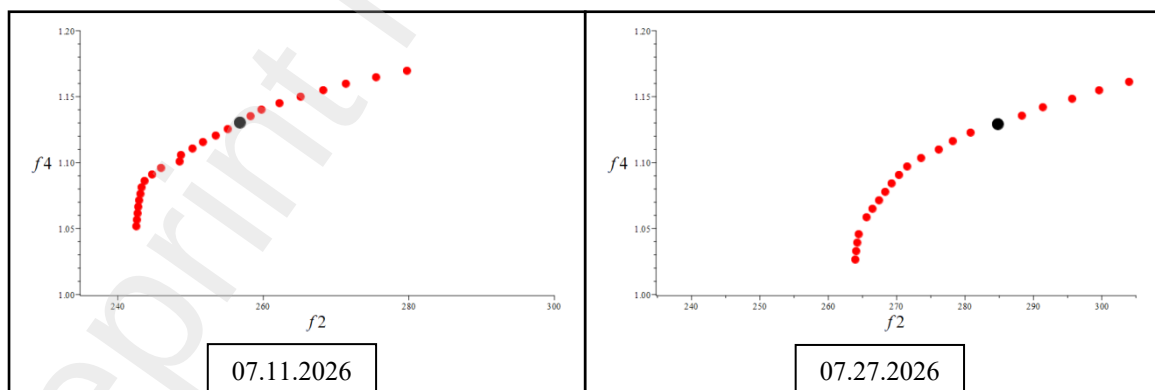
**Fig. 17.** The approximation of Pareto frontiers of MOP3 on two selected days in Case Study 2.

Fig. 17 and the results presented in Table 12 indicate that, in the third multi-objective optimization model, a larger BESS capacity is proposed for July 11 compared to July 27 to achieve a reliability level of approximately 1.13, while the required output power remains nearly the same for both Pareto-optimal solutions.

**Table 12.** Main characteristics of the selected Pareto-optimal solution of MOP 3 for Case Study 2.

| Date       | BESS           |            | System reliability |
|------------|----------------|------------|--------------------|
|            | Capacity (kWh) | Power (kW) |                    |
| 07.11.2026 | 85.31          | 29.74      | 1.13               |
| 07.27.2026 | 57.68          | 28.85      | 1.13               |

Fig. 18 presents the approximated Pareto frontiers generated from MOP4.



**Fig. 18.** The approximation of Pareto frontiers of MOP 4 on two selected days in Case Study 2.

The details of the two selected Pareto-optimal solutions are summarized in Table 13.

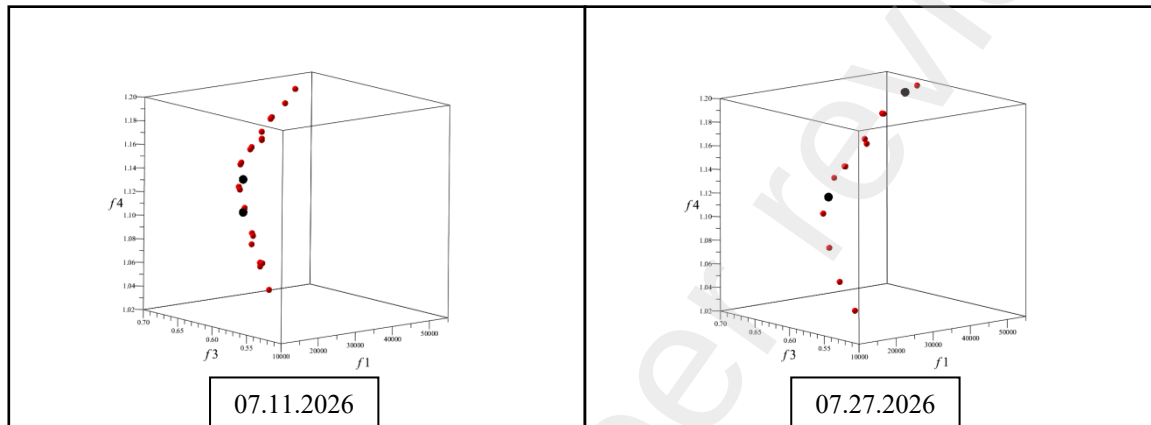
**Table 13.** Main characteristics of the selected Pareto-optimal solution of MOP 4 for Case Study 2.

| Date | BESS | System reliability |
|------|------|--------------------|
|------|------|--------------------|

|            | Capacity (kWh) | Power (kW) |      |
|------------|----------------|------------|------|
| 07.11.2026 | 106.61         | 33.01      | 1.13 |
| 07.27.2026 | 67.38          | 27.47      | 1.13 |

The results presented in Table 13 indicate that MOP4 recommends a larger BESS capacity for July 11 compared to July 27 to achieve a system reliability of approximately 1.13.

The approximation of the Pareto frontiers of the three-objective optimization model, MOP 5, for July 11 and 27 is shown in Fig. 19.



**Fig. 19.** The approximation of Pareto frontiers of MOP 5 on two selected days in Case Study 2.

Two Pareto-optimal solutions are selected from each model for further analysis: one solution with approximately equal system reliability values on both days, and another with approximately equal self-sufficiency ratios on both days. Table 14 provides the details of selected Pareto-optimal solutions.

**Table 14.** Main characteristics of the selected Pareto-optimal solution of MOP 5 for Case Study 2.

| Date       | BESS           |            | Self-sufficiency ratio (%) | System reliability |
|------------|----------------|------------|----------------------------|--------------------|
|            | Capacity (kWh) | Power (kW) |                            |                    |
| 07.11.2026 | 85.31          | 17.39      | 63.26%                     | 1.10               |
|            | 115.18         | 48.33      | 67.55%                     | 1.13               |
| 07.27.2026 | 150.00         | 42.28      | 63.77%                     | 1.19               |
|            | 49.05          | 27.48      | 59.40%                     | 1.13               |

The results in Table 14 indicate that, to achieve an approximately equal self-sufficiency ratio, MOP5 recommends a BESS with greater capacity and power for July 27 compared to July 11, along with higher system reliability. Conversely, a larger BESS capacity and power are required for July 11 to attain approximately equal system reliability, resulting in a higher self-sufficiency ratio.

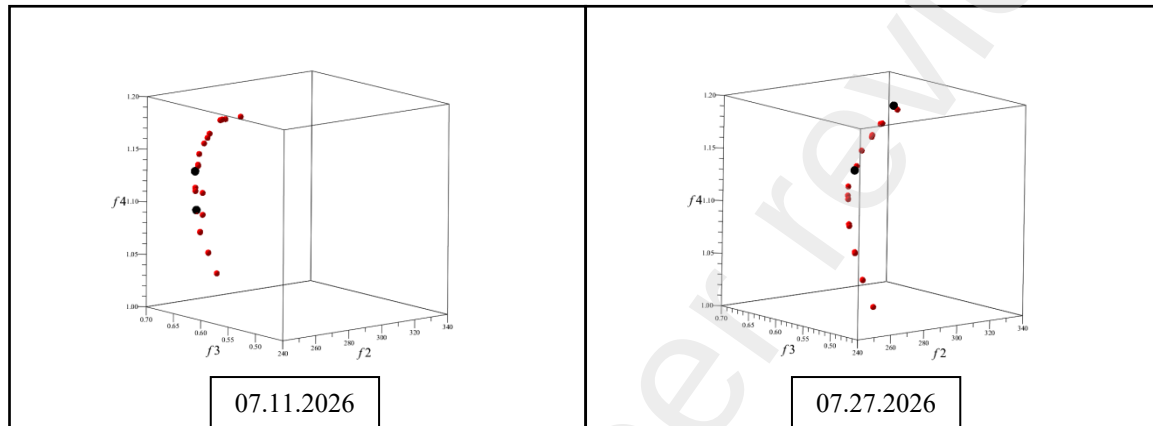
Fig. 20 illustrates the approximated Pareto frontier for MOP6, and the details of the selected Pareto-optimal points are provided in Table 15.

**Table 15.** Main characteristics of the selected Pareto-optimal solution of MOP 6 for Case Study 2.

| Date | BESS           |            | Self-sufficiency ratio (%) | System reliability |
|------|----------------|------------|----------------------------|--------------------|
|      | Capacity (kWh) | Power (kW) |                            |                    |

|            |        |       |        |      |
|------------|--------|-------|--------|------|
| 07.11.2026 | 106.61 | 32.24 | 63.68% | 1.09 |
|            | 122.00 | 48.94 | 66.32% | 1.13 |
| 07.27.2026 | 150.00 | 50.00 | 63.32% | 1.18 |
|            | 68.58  | 29.93 | 60.01% | 1.13 |

A similar pattern can also be observed in the results of MOP6. Comparison of the results from the three-objective models, MOP5 and MOP6, on the same days reveals that imposing DOD constraints in MOP6 necessitates a BESS with a larger energy capacity to achieve comparable outcomes.

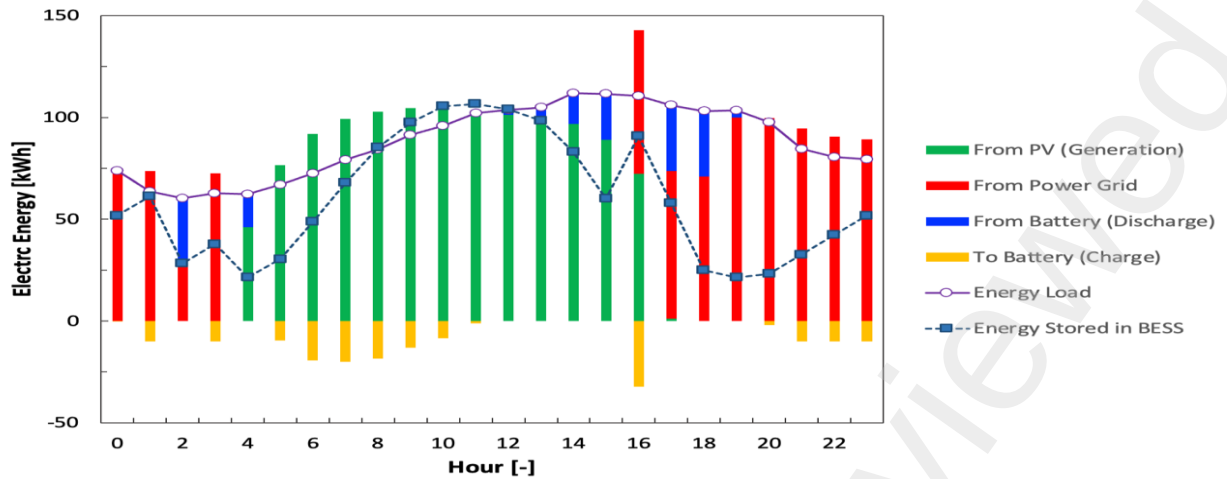


**Fig. 20.** The approximation of Pareto frontiers of MOP 6 on two selected days in Case Study 2.

**Case Study 3.** This experiment evaluates the BESS scheduling algorithm proposed in Section 3.2. For a concrete assessment, forecasted data for energy consumption and PV generation on July 11, 2026, from Case Study 2 are utilized, along with one of the Pareto-optimal solutions from the MOP6 model presented in Table 15. This selected Pareto solution recommended a BESS with an energy capacity of 106.61 kWh and a charging power of 32.24 kW. In addition to the forecasted energy consumption and PV generation data, Fig. 21 illustrates the operational characteristics of the recommended BESS, including charging and discharging power profiles and the state of energy storage (SoC) over a 24-hour period, as derived from the MOP6 optimization results.

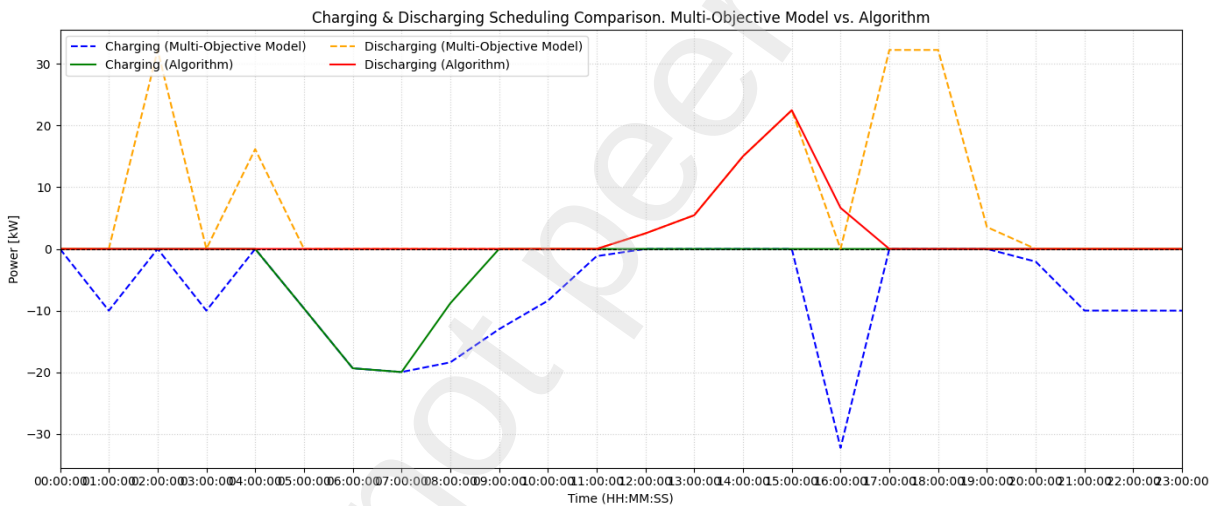
The selected Pareto-optimal solution specifies both the initial and final state of charge (SoC) at 51.75 kWh. To ensure consistency with this configuration, the same SoC condition was incorporated into our proposed scheduling algorithm.





**Fig. 21.** System profile and BESS operating status based on a selected Pareto-optimal solution from the MOP6 model for July 11, 2026.

Fig. 22 provides a comparative analysis of the BESS charge and discharge profiles generated by the MOP6 model and the proposed algorithm.



**Fig. 22.** Comparison of BESS charge and discharge profiles derived from the MOP6 and the proposed scheduling algorithm.

While the multi-objective optimization model provides an optimal scheduling plan based on predefined system inputs and objectives, Fig. 22 highlights the dynamic behavior of the proposed algorithm in real-time scenarios. The algorithm charges the BESS during periods of high PV generation or low electricity prices and discharges during peak demand, adapting to changing conditions. In contrast, the optimization model may pre-schedule early morning charging to ensure a sufficient state of charge (SoC), while the algorithm postpones such actions unless predefined scoring criteria are satisfied. The optimization model achieves precision through fine-tuned, hour-by-hour power adjustments, whereas the algorithm relies on discrete decisions guided by forecast-based scores. This allows the algorithm to be more responsive to real-time fluctuations or unforeseen events, unlike the optimization model, which requires re-solving when system conditions change. Moreover, the algorithm enhances operational transparency by attaching explicit rationales to each decision (e.g., charging due to surplus PV or favorable pricing). In comparison, the optimization model, though accurate, may



lack immediate interpretability without additional analysis. This computational experiment demonstrates that the forecast-driven BESS control algorithm effectively manages energy storage while adhering to system constraints. Compared to the optimization-based approach, the algorithm provides a more adaptable, interpretable, and practical solution for real-world smart grid applications.

## 5. Conclusion

This study proposes several multi-objective optimization models for determining the optimal sizing and operation of Battery Energy Storage Systems (BESS) in a grid-connected microgrid (MG) with photovoltaic (PV) energy generation while considering both economic and technical criteria. Six different multi-objective optimization models, including two and three-objective functions, were developed to provide flexible decision-making frameworks. By introducing a set of Pareto-optimal solutions, the proposed models enable decision-makers to select the most appropriate option based on broader management criteria rather than relying solely on a single solution.

Artificial intelligence prediction techniques have been utilized to enhance the accuracy of energy generation and consumption forecasts, as well as to support multi-objective optimization models. Additionally, a forecast-driven algorithm was developed to manage BESS charging and discharging operations, utilizing predicted PV output and energy consumption to optimize storage behavior over a 24-hour horizon dynamically.

Computational experiments confirmed the effectiveness of the proposed approach, highlighting its potential to optimize energy storage operations and improve the performance and reliability of renewable energy-based microgrids.

## CRedit authorship contribution statement

**Behzad Pirouz:** Conceptualization, Data curation, Investigation, Software, Writing – original draft. **Francesca Guerriero:** Conceptualization, Formal analysis, Methodology, Supervision, Validation, Funding acquisition, Writing – review and editing. **Angelo Algieri:** Validation, Writing – review and editing.

## Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## ACKNOWLEDGMENT

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## Data availability

Data will be made available on request.

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